(1.1) Systems of linear equations

Ex 1: $x + 2y = 3$ is linear system (1 equation, 2 variables)

Ex 2: $2x - y + z = 7$, $x + 3y - z = 1$ is linear system (2 equations, 3 variables)

Ex 3: $x_1 + 3x_2 = 1$, $2x_1 - x_2 = 0$ linear system, 2 eqns, 2 var.

Ex 4: $x_1 - 3x_3 = 2$, $2x_1 - 5x_2 = 1$, $x_2 - x_3 = 0$ linear system, 3 eqns, 3 variables

Ex: $x^2 + 2y = 1$, $2x - y = 3$ } is not linear system

Ex: $x + 3y = 1$, $2x + \ln y = 3$ } is not linear system

Ex: $x - y + \frac{1}{z} = 0$, $\sqrt{x} - 3z = -2$ } is not linear system

Ex: $xy = 2$, $2x - 3y = 1$ } is not linear system

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Definition (linear system)

General linear system consists of (m) equations in (n) unknown variables can be written as

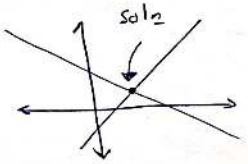
$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

* The solution of the linear system is the set of numbers that satisfy the linear system

Types of solutions for linear system

① unique solution (one solution)

if number of independent equations = number of variables



Ex: $x - y = 1$

$2x + y = 6$

$x = \frac{7}{3}$, $y = \frac{4}{3}$.

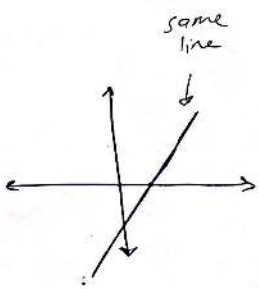
or $(x, y) = (\frac{7}{3}, \frac{4}{3})$

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(2) infinite number of solutions

(no. of indep. eqns < no. of variables)

$$2x - y = 3$$
$$4x - 2y = 6$$



$$(x, y) = \left(\frac{3}{2} + \frac{1}{2}t, t\right) \quad t \in \mathbb{R}$$

$t = 1 \quad (x, y) = (2, 1)$ is soln

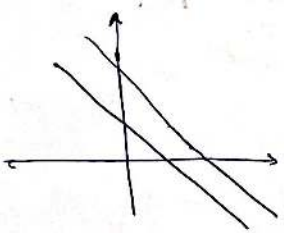
$t = 2 \quad = \left(\frac{5}{2}, 2\right)$ " "

$t = -1 \quad = (1, -1)$ " "

⋮ ⋮

(3) No solution

if we have contradiction



$$x + y = 4$$
$$3x + 3y = 6$$

$-3 \text{ eq}^n ① + \text{eq}^n ② \longrightarrow 0 = -6$!!

* If the system has solution (unique or infinite) then it is consistent

* If the system has No soln then it is inconsistent.

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(1.2) Gaussian Elimination

For the following $m \times n$ linear system

(m: no. of eqns; n: no. of variables)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

} m equations, n variables

Define the augmented matrix to be

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

is matrix of size $m \times (n+1)$

(m: no. of rows = no. of eqns; $n+1$: no. of columns = no. of var. + 1)

Ex: Find the augmented matrix for the linear system

① $3x_1 - 2x_2 = -1$
$4x_1 + 5x_2 = 3$
$7x_1 + 3x_2 = 2$

→ The augmented matrix is $\begin{bmatrix} 3 & -2 & -1 \\ 4 & 5 & 3 \\ 7 & 3 & 2 \end{bmatrix}$

② $x - 3z = -2$
$2y + z = 1$
$7x - 2y - 3 = 0 \rightarrow 7x - 2y = 3$

→ The augmented matrix $= \begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 2 & 1 & 1 \\ 7 & -2 & 0 & 3 \end{bmatrix}$

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Ex: Given the augmented matrix write the linear system

① $\begin{bmatrix} 2 & -1 & 3 & 4 \\ 3 & 0 & 2 & 5 \end{bmatrix} \rightarrow \begin{aligned} 2x - y + 3z &= 4 \\ 3x + 2z &= 5 \end{aligned}$

② $\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \rightarrow \begin{aligned} x &= 2 \\ y &= -1 \\ z &= 3 \end{aligned}$ or $\begin{aligned} x_1 &= 2 \\ x_2 &= -1 \\ x_3 &= 3 \end{aligned}$

③ $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{aligned} x &= -1 \\ y &= 2 \\ 0 &= 0 \checkmark \end{aligned}$

④ $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{aligned} x &= -1 \\ y &= 2 \\ z &= 1 \\ 0 &= 3 \ !! \end{aligned}$ No soln

⑤ $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{aligned} x + 2z &= 1 \\ y - z &= 1 \\ 0 &= 0 \end{aligned} \rightarrow \begin{aligned} x &= 1 - 2z \\ y &= 1 + z \\ z &= z \end{aligned}$ infinite no. of solns.

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Reduced Row echelon form

Definition:

A matrix is in reduced row echelon form if the matrix satisfied:

① The first nonzero entry in each row is 1 we call it the leading 1 (except the zero row).

② zero row(s) if exist grouped at the bottom of the matrix.

③ Each two successive rows the leading 1 in the lower row occur further to the right the leading 1 in the higher row.

④ Each column that contains leading 1 has zeros everywhere else in that column.

If the matrix satisfied the first three properties, then it is said to be row echelon form (R.E)

If it satisfied the four properties it is reduced row echelon form. (R.R.E).

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Examples –

Determine whether the matrices R.R.E or R.E or not.

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

R.R.E

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Not R.E

$$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

not R.E

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

R.R.E

$$\begin{bmatrix} 1 & 3 & 8 & 1 \\ 0 & 1 & -2 & 3 \end{bmatrix}$$

R.E

$$\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

not R.E

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

R.R.E

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

R.E

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

R.E

$$\begin{bmatrix} 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

R.R.E

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* If the augmented matrix transformed to R.R.E matrix, then we can find the type of the solution.

linear system $\longrightarrow$ augmented matrix $\xrightarrow{\text{by elementary row operations}}$ R.R.E matrix

Ex: Find the solution for the following R.R.E matrices.

① $\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{matrix} x=2 \\ y=-3 \\ z=0 \end{matrix}$ $(x,y,z)=(2,-3,0)$ unique solution

② $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{matrix} x=0 \\ y+2z=0 \\ 0=1 \text{ !! contradiction} \end{matrix}$

$\rightarrow$ No solution

③ $\begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \left.\begin{matrix} x+3z=1 \\ y-4z=2 \end{matrix}\right\} \rightarrow \begin{matrix} x=1-3z \\ y=2+4z \end{matrix}$

$0=0$

$\rightarrow (x,y,z)=(1-3z, 2+4z, z)$ $z\in\mathbb{R}$

or $=(1-3t, 2+4t, t)$, $t\in\mathbb{R}$

④ $\begin{bmatrix} 1 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $x-5y+z=4 \rightarrow x=4+5y-z$

$0=0$

$(x,y,z)=(4+5y-z, y, z)$, $y,z\in\mathbb{R}$

$=(4+5t-s, t, s)$ $t,s\in\mathbb{R}$

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(1.2) Gauss-Jordan elimination

To solve linear system by Gauss-Jordan elimination

1) Given the linear system
2) Write the augmented matrix.
3) Using elementary row operations transform the matrix to R.R.E.
4) Write the sol^{n} (if exist) by determining the sol^{n} (unique or infinite or No sol^{n}).

Ex: Solve the linear system by G-J elimination

1) $2x - y = 3$
$x + 2y = 4$

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 2 & -1 & 3 \end{bmatrix}^{R_1 = R_2}_{R_2 = R_1} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -5 & -5 \end{bmatrix}^{-2R_1 + R_2}$$

$$= \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 1 \end{bmatrix}^{\frac{R_2}{-5}} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}^{-2R_2 + R_1}$$

$x = 2, \; y = 1$ or $(x, y) = (2, 1)$

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2) $y - z = -4$

$2x - y + z = 8$

$3x - z = 3$

$$\begin{bmatrix} 0 & 1 & -1 & -4 \\ 2 & -1 & 1 & 8 \\ 3 & 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & -1 & 3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & -1 & -4 \end{bmatrix} R_1 - R_2$$

$$= \begin{bmatrix} 1 & 1 & -2 & -5 \\ 0 & -3 & 5 & 18 \\ 0 & 1 & -1 & -4 \end{bmatrix} -2R_1 + R_2 = \begin{bmatrix} 1 & 1 & -2 & -5 \\ 0 & 1 & -1 & -4 \\ 0 & -3 & 5 & 18 \end{bmatrix} \begin{matrix} R_2 \leftrightarrow R_3 \\ R_3 \leftrightarrow R_2 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 2 & 6 \end{bmatrix} \begin{matrix} R_1 - R_2 \\ \\ 3R_2 + R_3 \end{matrix} = \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 1 & 3 \end{bmatrix} R_3/2$$

$$= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} R_1 + R_3 \\ R_2 + R_3 \\ \\ \end{matrix}$$

$(x, y, z) = (2, -1, 3)$ unique sol^n

3) $x_1 - x_2 + 2x_3 = 1$

$x_1 + 2x_2 - x_3 = 2$

$$\rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 \\ 1 & 2 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 1 \end{bmatrix} R_2 - R_1 = \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 1/3 \end{bmatrix} R_2/3$$

$$= \begin{bmatrix} 1 & 0 & 1 & 4/3 \\ 0 & 1 & -1 & 1/3 \end{bmatrix} R_1 + R_2$$

$(x_1, x_2, x_3) = \left(\frac{4}{3} - t, \frac{1}{3} + t, t\right)$, $t \in \mathbb{R}$

infinite no. of sol^{n}s.

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4) $2x - 2y + 4z = -6$

$2x + y = 4$

$3x + 2z = 7$

$$\begin{bmatrix} 2 & -2 & 4 & -6 \\ 2 & 1 & 0 & 4 \\ 3 & 0 & 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 2 & 1 & 0 & 4 \\ 3 & 0 & 2 & 7 \end{bmatrix} R_1/2$$

$$= \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 3 & -4 & 10 \\ 0 & 3 & -4 & 16 \end{bmatrix} \begin{matrix} \\ -2R_1 + R_2 \\ -3R_1 + R_3 \end{matrix}$$

$$= \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 3 & -4 & 10 \\ 0 & 0 & 0 & 6 \end{bmatrix} R_3 - R_2$$

$\rightsquigarrow$ transform to eq$=$ $0 = 6$!!

$\rightarrow$ No soln.

Ex: Find all values of a such that the linear system $x + y = 3$, $x + (a^2 - 8)y = a$ has
a) No soln
b) unique soln
c) infinite soln

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & a^2-8 & a \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & a^2-9 & a-3 \end{bmatrix} R_2 - R_1$$

a) No soln $\quad a = -3$

b) unique soln $\quad a \neq \pm 3$

c) infinite soln $\quad a = 3$

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Homogeneous linear system

A linear system of the form

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0$$

is said to be homogeneous system

Every homog. linear system is consistent

If the soln → unique soln $x_1 = x_2 = \cdots = x_n = 0$ and called trivial soln

→ infinite no. of soln and called nontrivial soln

Ex: Solve the homog. systems

1) $x_1 + 2x_2 - x_3 = 0$
$2x_1 - x_2 + 3x_3 = 0$
$3x_1 + 6x_2 - 3x_3 = 0$

soln
$(x_1, x_2, x_3) = (-t, t, t)$, $t \in \mathbb{R}$
nontrivial soln

2) $x_1 + 3x_2 - 2x_3 + 2x_5 = 0$
$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = 0$
$5x_3 + 10x_4 + 15x_6 = 0$
$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 0$

$$\rightsquigarrow \begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$(x_1, x_2, x_3, x_4, x_5, x_6) = (-3t - 4r - 2s, t, -2r, r, s, 0)$, $t, r, s \in \mathbb{R}$

3) $5x - 2y + 3z = 0$
$2x - y + z = 0$
$-3x + z = 0$

$(x, y, z) = (0, 0, 0)$ trivial soln.

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* If the number of eqns < no. of variables, then soln of homog. system has nontrivial soln.

* If the homog. system $AX=0$ has only trivial soln, then the linear system $AX=B$ has unique soln.

Ex: Solve $2x-y=4$ and $2x-y=0$
$3x+2y=1$ $3x+2y=0$

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(1.3) Matrices and Matrix operations

Def: (matrix)

a matrix is a rectangular array of numbers.

These numbers are called entries

The size of the matrix is $m \times n$ means that it has m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

ith row is $[a_{i1}\ a_{i2}\ \cdots\ a_{in}]$

jth column is $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$

entry in the ith row and jth column is a_{ij}

Def: (Square matrix)

The matrix A is called square matrix if $n = m$ (no. of rows = no. of columns).

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

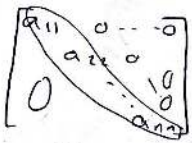
A is $n \times n$ matrix

→ is the main diagonal.

Ex: $A = \begin{bmatrix} 2 & 1 \\ 3 & -4 \end{bmatrix}$ 2×2 matrix , $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 7 & 1 & 2 \end{bmatrix}$ 3×3 matrix

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Def: (Diagonal matrix)

matrix A is called diagonal matrix if $a_{ij} = 0$ when $i \neq j$ that is $A =$ 

Ex: $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

Def: upper triangular and lower triangular matrices $A = \begin{bmatrix} a_{11} & * \\ 0 & a_{nn} \end{bmatrix}$ upper, $B = \begin{bmatrix} a_{11} & 0 \\ * & a_{nn} \end{bmatrix}$ lower

Def: (Equality of matrices)

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are two matrices of the same size and $a_{ij} = b_{ij}$ for all i and j then $A = B$

Ex: $A = \begin{bmatrix} 2 & 1 \\ 3 & x \end{bmatrix}$ $B = \begin{bmatrix} y & 1 \\ 3 & 5 \end{bmatrix}$

If $A = B$ find x and y

* Matrix operations

Matrix addition and Substraction

If $A = [a_{ij}]$ $m \times n$ matrix and $B = [b_{ij}]$ $m \times n$ matrix

then $A \pm B = [a_{ij} \pm b_{ij}]$ is $m \times n$ matrix

Ex: $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 7 \\ 2 & 0 \end{bmatrix}$ find $A+B$, $B-A$

Scalar multiplication

If $A = [a_{ij}]$ $m \times n$ matrix and $c \in \mathbb{R}$ is scalar (constant)

then $cA = [ca_{ij}]$ $m \times n$ matrix.

Ex: $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & -2 \end{bmatrix}$ find $3A$, $-2A$

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Def. (identity matrix)

Identity matrix is diagonal matrix where entries of the diagonal are 1's and denoted by I_n

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, ...

Matrix multiplication

If $A = [a_{ij}]$ is $m \times r$ matrix and $B = [b_{ij}]$ $r \times n$ matrix, then the product AB is $m \times n$ matrix

The entry in the ith row and jth column was found by multiplying the corresponding entries in the ith row in A and jth column in B

$$c_{ij} = [a_{i1}\ a_{i2}\ \cdots\ a_{ir}] \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{rj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ir}b_{rj}$$

Ex: $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix}$ 2×3, $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 3 \\ -1 & -2 & 1 \end{bmatrix}$ 3×3

find AB

Ex: $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 4 & 3 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ find AB

* The product is defined for matrices only if the number of columns in the 1st matrix = number of rows in the second matrix.

A B $\rightarrow$ AB is $m \times n$
$m \times r$ $r \times n$

Ex: A is 3×3, B is 2×3, C is 2×2
AB undefined, BA defined, CB defined

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* note that if AB and BA is defined, it is not necessary that $AB = BA$.

Ex: $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}_{2\times 3}$, $B = \begin{bmatrix} 1 & 3 \\ 0 & -1 \\ 4 & 2 \end{bmatrix}_{3\times 2}$

find AB and BA.

Ex: $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ Find AB, BA

* If $I_n = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$ is $n\times n$ identity matrix and $I_m = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$ is $m\times m$ " "

$A = [a_{ij}]$ is $m\times n$ matrix, then

$I_m A = A I_n = A$.

Ex: $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}_{2\times 3}$ find $I_2 A$ and $A I_3$

* $A^n = A.A...A$ n is integer and A is square matrix.

Ex: $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ find A^3.

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Properties of matrix addition ~~and scalar mult.~~

(1) $A+B = B+A$.

(2) $A+(B+C) = (A+B)+C$

(3) $A+0 = 0+A = A$
where 0 is the zero matrix with the same size of A

(4) $A+(-A) = (-A)+A = 0$

Note zero matrix 0 is matrix all entries are zeros

Properties of scalar mult. $r,s \in \mathbb{R}$
A,B matrices of same size

(1) $r(sA) = rsA$

(2) $(r\pm s)A = rA \pm sA$

(3) $r(A\pm B) = rA \pm rB$

(4) $A(rB) = (rA)B = r(AB)$

Properties for multiplication

(1) $A(BC) = (AB)C$

(2) $A(B\pm C) = AB \pm AC$

(3) $(B\pm C)A = BA \pm CA$

(4) $a(AB) = (aA)B = A(aB)$

* No cancelation in product (not all cases)

~~Ex:~~ If $AB = AC$ that does not mean $B = C$

Ex: $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$

$AB = AC$ but $B \neq C$

~~Ex:~~ * In numbers if $ab = ac$, then $b = c$.
in matrices No.

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* zero product with non-zero factors

in numbers if $ab=0 \rightarrow a=0$ or $b=0$

not in matrices

Ex: $A=\begin{bmatrix}0 & 1\\ 0 & 2\end{bmatrix}$, $B=\begin{bmatrix}3 & 7\\ 0 & 0\end{bmatrix}$

$AB=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$ but $A\neq 0$, $B\neq 0$

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Transpose of a matrix

Def:
If $A=[a_{ij}]$ is $m \times n$ matrix, then $A^t=[a_{ji}]$ is $n \times m$ matr.x

Ex: $A=\begin{bmatrix}1 & -1 & 3\\ 2 & 1 & 0\\ 5 & 1 & 4\end{bmatrix}$, $B=\begin{bmatrix}2 & 1 & 5\\ -1 & 0 & 1\end{bmatrix}$, $C=[1 \quad 3 \quad 5]$

Properties:
① $(A^t)^t=A$ ② $(A \pm B)^t=A^t \pm B^t$ ③ $(AB)^t=B^tA^t$
④ $(cA)^t=cA^t$.

Def: symmetric matrix
A matrix $A=[a_{ij}]$ $n \times n$ is called symmetric matrix if $A^t=A$

Ex: $A=\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$, $B=\begin{bmatrix}1 & -1 & 2\\ -1 & 3 & 5\\ 2 & 5 & 7\end{bmatrix}$, $C=\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & 5\end{bmatrix}$

$D=\begin{bmatrix}4 & 2\\ 2 & -1\end{bmatrix}$

Def: Trace of A matrix
If $A=[a_{ij}]$ is $n \times n$ matrix, then trace A denoted by trA is the sum of the main diagonal entries.

$trA = a_{11}+a_{22}+\cdots+a_{nn}$

Ex: $A=\begin{bmatrix}-1 & 2\\ 0 & 3\end{bmatrix}$ find trA, $B=\begin{bmatrix}-1 & 2 & 7 & 6\\ 3 & 5 & -8 & 4\\ 1 & 2 & 7 & -3\\ 4 & -2 & 1 & 0\end{bmatrix}$ find trB

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* If A and B are symm. then

(1) A^b is symm

(2) $A \pm B$ is symm

(3) kA is symm

(4) If A invert. symm, then A^{-1} symm

(5) AA^t, A^tA is symm, for any matrix A

Prove all.

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Matrix form of linear system

Consider the linear system of m equations and n variables

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}\right\} \quad m \times n \text{ linear system}$$

Define the following matrices.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \text{ is } m \times n \text{ matrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ is } n \times 1 \text{ matrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \text{ is } m \times 1 \text{ matrix}$$

The system is equivalent to the matrix form

$$AX = B$$

Ex: Find the matrix form for the system

$$\begin{aligned} 2x - 3y + 2z &= 1 \\ x - 2z &= -3 \\ -x + y &= 4 \end{aligned}$$

Ex: Find the system $AX = B$ where

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

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The matrix $[A:B]$ is called the augmented matrix.

Ex: Find the augmented matrix for the linear system

$$2x + 3y - 4z = 5$$
$$-2x + z = 7$$
$$3x + 2y + 2z = 3$$

Ex: The matrix

$$\begin{bmatrix} 2 & -1 & 3 & 4 \\ 3 & 0 & 2 & 5 \end{bmatrix}$$ is augmented matrix

Find the corresponding linear system

Ex: Find the linear system for the augmented matrix

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Ex: Find the linear system

$$\begin{bmatrix} 1 & 0 & -5 & 0 & 2 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

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1.4 Inverse of a Matrix

NOTE: inverse only for square matrices.
in numbers $a^{-1} = \frac{1}{a}$ thas not in matrices.

Def:
If A is $n \times n$ matrix, then A is called invertible (nonsingular) if there exist $n \times n$ matrix denoted by A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n$$

If A^{-1} does not exist, then A is called singular matrix (noninvertible).

* The inverse of any square matrix is unique

Ex. $A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

$AB = BA = I_2$ then $B = A^{-1}$ (and $A = B^{-1}$)

Properties

① If A invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$

② If A and B are invertible, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

③ If A is invertible, then A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$

④ $A^{-n} = (A^{-1})^n = A^{-1} \cdot A^{-1} \cdots A^{-1}$ n times $(A^2)^{-1} = A^{-1} \cdot A^{-1}$

⑤ $(cA)^{-1} = \frac{1}{c}A^{-1}$, $c \neq 0$

* note $(A+B)^{-1} \neq A^{-1} + B^{-1}$, $(AB)^{-1} \neq A^{-1}B^{-1}$

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inverse for any 2×2 matrix

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc \neq 0$, then

A is invertible (A^{-1} exist) and

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(note $ad-bc$ is the determinant of A
$= \det A = |A|$)

Ex: $A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 3 & +6 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

find A^{-1}, B^{-1}, C^{-1}, $(AB)^{-1}$, $(2A)^{-1}$, $(A^t)^{-1}$, $(B^{-1})^{-1}$

$(B^2)^{-1}$

Ex: $A^{-1} = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix}$ find A

* another way to find A^{-1} take $[A : I_n]$ $\xrightarrow{\text{elementary row operations}}$ $[I_n : A^{-1}]$

Solve linear system using inverse

If $AX = B$ linear system and

① A is $n \times n$ matrix ② A is invertible, then $X = A^{-1}B$

Ex: solve the linear system

$x + y = 3$
$2x + y = -4$.

using inverse.

② Solve $x + 2y + 3z = 4$
$2x + 5y + 3z = 5$
$x + 8z = 9$

using inverse

$\left(A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}, X = \begin{bmatrix} ! \\ ! \end{bmatrix} \right.$

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Chapter II

Determinants

* Determinants only for square matrices
* Determinant of any square matrix is real number
* If $\det A \neq 0$, then A^{-1} exist (A is invertible
  If $\det A = 0$, then A^{-1} does not exist (A is non invert. (singular))
* $\det A = |A|$

* Determinant for 2x2 matrices

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ (+ on $a_{11}a_{22}$, − on $a_{12}a_{21}$)

$\det A = a_{11}a_{22} - a_{12}a_{21}$

Ex: $A = \begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 5 \\ 2 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 3 & -6 \\ 2 & -4 \end{bmatrix}$ find $\det A$, $\det B$, $\det C$

* Definition (Minor)

Let $A = [a_{ij}]$ be $n \times n$ matrix, the minor of a_{ij} denoted by M_{ij} is the determinant of the submatrix that remains after deleting ith row and jth column.

The cofactor of a_{ij} denoted by $C_{ij} = (-1)^{i+j} M_{ij}$

Ex: $A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$ Find M_{12}, M_{31}, C_{22}

~~Find~~ matrix of minors is $= \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}$

~~Find~~ " " cofactors is $\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$.

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* Determinant for 3×3 matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Choose any row or any column

row 1 $\det A = a_{11}M_{11} + a_{12}M_{12} + a_{13}C_{13} = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$

or column 2 $\det A = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}$

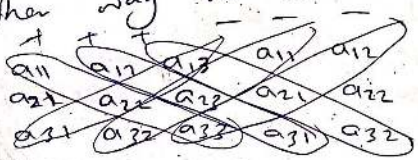
$= -a_{12}M_{12} + a_{22}M_{22} - a_{32}M_{32}$

row 3 $\det A = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$

$= a_{31}M_{31} - a_{32}M_{32} + a_{33}M_{33}$

* the detA will be the same number by using any row or any column

Another way for 3×3 matrix



$\det A = +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$

$- a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$

Ex. $A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$ find $\det A$ $(= -1)$

$B = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$ find $\det B$ (Hw)

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Finding inverse of a matrix

If A is $n \times n$ matrix and $\det A \neq 0$, then A^{-1} exist (A is invertible) (nonsingular)

Def: Adjoint of a matrix

let $M(A) = \begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1n} \\ M_{21} & \cdots & & M_{2n} \\ M_{n1} & \cdots & & M_{nn} \end{bmatrix}$ be the matrix of minors

let $C(A) = \begin{bmatrix} +M_{11} & -M_{12} & \cdots & (-1)^{1+n} M_{1n} \\ \vdots & & & \\ (-1)^{n+1} M_{n1} & \cdots & & M_{nn} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ \vdots & & & \\ C_{n1} & \cdots & & C_{nn} \end{bmatrix}$

be the matrix of Cofactors, then Adjoint of A is the transpose of the matrix of cofactors

$$Adj(A) = (C(A))^t$$

Theorem:

If A is $n \times n$ matrix, $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} Adj(A)$$

Ex: $A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$ find A^{-1} $\left(A^{-1} = \frac{1}{64} \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix} \right)$

$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ find B^{-1} $\left(B^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \right)$

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Theorem:
If A is $n \times n$ matrix, then the following statements are equivalent

1. A is invertible
2. $AX = 0$ has only the trivial solution
3. $AX = B$ has unique solution $(X = A^{-1}B)$ (consistent)
4. A is row equivalent to I_n.
5. $\det A \neq 0$

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* Theorem

Let A be $n \times n$ matrix. For each $1 \le i \le n$

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

and for each $1 \le j \le n$

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

Ex: $A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}$ Find $\det A$ $(= -6)$

* Properties of determinant

(1) For any $n \times n$ matrix $\det A = \det A^t$.

(2) If we interchange two rows (or two columns) the determinant multiplied by -1.

Ex $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 4 \\ 5 & 0 & 1 \end{bmatrix}$ $\det A = 45$, $B = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 1 \\ 1 & 3 & 4 \end{bmatrix}$, $\det B = -45$

Ex: $A = \begin{bmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$, $\det A = -2$, $B = \begin{bmatrix} a & c & b \\ a_1 & c_1 & b_1 \\ a_2 & c_2 & b_2 \end{bmatrix}$, $C = \begin{bmatrix} a_2 & c_2 & b_2 \\ a_1 & c_1 & b_1 \\ a & c & b \end{bmatrix}$

Find $\det B$, $\det C$.

(3) If any two rows (two columns) in an $n \times n$ matrix are equal, then $\det A = 0$

Ex. Find $\det \begin{bmatrix} 1 & -1 & 3 & 3 \\ 2 & 0 & 1 & 1 \\ -1 & 0 & -1 & -1 \\ 2 & 5 & 4 & 4 \end{bmatrix}$, $\det \begin{bmatrix} a & b & c \\ a_2 & b_2 & c_2 \\ a & b & c \end{bmatrix}$

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④ If a matrix contains zero row (zero column), then $\det A = 0$

Ex: $A = \begin{bmatrix} 1 & -1 & 5 \\ 0 & 0 & 0 \\ 1 & 2 & 4 \end{bmatrix}$ $\det A = 0$

⑤ If we make the following change to a row in A $R_i = R_i + cR_j$, then the determinant does not change (do not multiply the row by constant)

Ex: $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & -3 & 6 \end{bmatrix}_{R_3 = R_3 - 2R_1}$

, then $|A| = \begin{vmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & -3 & 6 \end{vmatrix} = 1(18+3) = 21$.

⑥ If we multiply one row (one column) by constant, then the determinant multiplied by this constant.

Ex: $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix}$

$\det B = 2 \det A$

Ex: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det A = -3$, $B = \begin{bmatrix} a & 3b \\ c & 3d \end{bmatrix}$, $C = \begin{bmatrix} 2a & 6b \\ c & 3d \end{bmatrix}$

$D = \begin{bmatrix} 2c & d \\ -2a & -b \end{bmatrix}$ find $\det B$, $\det C$, $\det D$.

* If A is $n \times n$ matrix, $c \in \mathbb{R}$

$\det cA = c^n \det A$

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⑦ If A is lower triangular matrix, or upper triangular matrix or diagonal matrix, then $\det A = a_{11} a_{22} \cdots a_{nn}$

$$A = \begin{bmatrix} a_{11} & & * \\ & \ddots & \\ 0 & & a_{nn} \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} a_{11} & & 0 \\ & \ddots & \\ * & & a_{nn} \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{bmatrix}$$

Ex: Find $\det \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -3 & 5 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ 1 & -3 & 2 & 4 & 0 \\ -1 & 1 & 0 & 0 & 3 \end{bmatrix}$

Ex: Use det properties to transform A to lower triangular matrix to find $\det A$.

$$A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 2 & 7 & 0 & 6 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -5 \end{bmatrix}$$

$$= \det \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 7 & 0 & 0 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -26 \end{bmatrix} \quad \begin{matrix} c_{13} = -3c_1 + c_3 \\ \\ \\ \\ \end{matrix} \qquad = -546$$

Ex: $A = \begin{bmatrix} 2 & -2 & -3 \\ 3 & 1 & +3 \\ 5 & 3 & 4 \end{bmatrix}$

$$\det A = \det \begin{bmatrix} 8 & 0 & +3 \\ 3 & 1 & -3 \\ -4 & 0 & 13 \end{bmatrix} \begin{matrix} 2R_2 + R_1 \\ \\ -3R_2 + R_3 \end{matrix} = -1(8 + 12) = -20$$

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Theorem:

For any $n \times n$ matrix A & B

$$\det AB = \det A \cdot \det B$$

Corollary 1:

$$\det A^n = (\det A)^n$$

Corollary 2:

$$\det A^{-1} = \frac{1}{\det A}$$

(Proof: $\det I = 1$

$AA^{-1} = I \rightarrow \det AA^{-1} = 1$

$\det A \cdot \det A^{-1} = 1$

$\rightarrow \det A^{-1} = \frac{1}{\det A}$)

Ex: $\det A = -3$, $\det B = 2$ For 2×2 matrices A, B

Find $\det A^2$, $\det 2A^{-1}$, $\det(3A)^{-1}$

$\det 2B^{-1}A$; $\det(AB)^{-1}$, $\det B^{-2}$, $\det(2A)^t$

Ex: Give an example to show that

$\det(A \pm B) \neq \det A \pm \det B$

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Ex: Solve the linear system

$$x - 2z = 2$$
$$2x + y + 3z = 1$$
$$4x + 2y + 5z = 3$$

using (a) Gauss-Jordan elimination
(b) by inverse.

Solve $x - 2z = 0$
$2x + y + 3z = 0$
$4x + 2y + 5z = 0$ using inverse

Theorem:
If A is $n \times n$ matrix, then the following are equivalent

(1) A is invertible.
(2) $AX = 0$ has only trivial sol^n.
(3) $AX = B$ has unique soln $X = A^{-1}B$.
(4) A is row equivalent to I_n.

* Note.
If the linear system is not square or if A is not invertible we can solve the linear system only by Gauss-Jordan

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* If A is diagonal matrix and $a_{ii} \neq 0$ for all i

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & & & \\ 0 & \cdots & & a_{nn} \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & & & \\ 0 & \cdots & & \frac{1}{a_{nn}} \end{bmatrix}$$

Ex: find A^{-1}, B^{-1}

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2/3 & 0 \\ 0 & 0 & -1/4 \end{bmatrix}$$

Ex: $A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$ find A^{-1}.

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Cramer's rule

If $AX = B$ is system of n-linear equations and n-variables such that $\det A \neq 0$, then

$$x_1 = \frac{\det A_1}{\det A}, \quad x_2 = \frac{\det A_2}{\det A}, \quad \dots, \quad x_n = \frac{\det A_n}{\det A}$$

where A_j is the matrix A when deleting jth column by $\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = B$

Ex: Solve by cramer's rule

$$2x + y = 3$$
$$x - 3y = 5$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\rightarrow A_1 = \begin{bmatrix} 3 & 1 \\ 5 & -3 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$$

$$x = \frac{\det A_1}{\det A} = \frac{-14}{-7} = 2, \quad y = \frac{\det A_2}{\det A} = \frac{7}{-7} = -1$$

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Ex: Solve

$$x_1 + 2x_2 + 3x_3 = 2$$
$$2x_1 + 5x_2 + 3x_3 = 0$$
$$x_1 + 8x_3 = 1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 2 & 2 & 3 \\ 0 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \\ 1 & 1 & 8 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$x_1 = \frac{\det A_1}{\det A} = \frac{71}{-1} = -71$$

$$x_2 = \frac{\det A_2}{\det A} = \frac{-23}{-1} = 23$$

$$x_3 = \frac{-9}{-1} = 9$$

Ex: Solve

$$x + y - z = 1$$
$$y + z = -3$$
$$x - z = 3$$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ -3 \\ 3 \end{bmatrix}$$

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$$x = \frac{\det A_1}{\det A} = \frac{\det \begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 3 & 0 & -1 \end{bmatrix}}{\det \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}}$$

$$= \frac{+3(2) - 1(4)}{+1(-1) + 1(2)} = \frac{2}{1} = 2$$

$$y = \frac{\det A_2}{\det A} = \frac{\det \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{bmatrix}}{1}$$

$$= \frac{1(0) + 1(-2)}{1} = -2$$

$$z = \frac{\det A_3}{\det A} = \frac{\det \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 1 & 0 & 3 \end{bmatrix}}{1} =$$

$$= \frac{1(3) + 1(-4)}{1} = -1$$

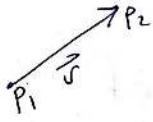
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Chapter III

Vectors

* vector is quantity that has magnitude and direction
* vector can be represented geometrically as directed line segments or arrows in 2-space or 3-space

$v = \overrightarrow{P_1P_2}$, P_1 is the initial point
P_2 is the terminal "



Def: n-vector is $n \times 1$ matrix $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

where $v_1, v_2, \dots, v_n$ are real numbers called components of v.

v also can be written as $v = (v_1, v_2, \dots, v_n)$

Def: let $u = (u_1, u_2, \dots, u_n)$, $v = (v_1, v_2, \dots, v_n)$

$u = v$ iff $u_i = v_i$ for all $1 \le i \le n$.

$0 = (0, 0, \dots, 0) \in \mathbb{R}^n$ is the zero vector.

$u \pm v = (u_1 \pm v_1, u_2 \pm v_2, \dots, u_n \pm v_n)$

$cu = (cu_1, cu_2, \dots, cu_n)$, c is constant

Properties: Let u, v and w vectors in $\mathbb{R}^n$ c, k are real numbers. Then

1. $u + v = v + u$
2. $u + (v + w) = (u + v) + w$
3. $u + 0 = 0 + u = u$
4. $u + (-u) = 0$
5. $c(ku) = (ck)u$
6. $c(k + v) = cu + cv$
7. $(c + k)u = cu + ku$
8. $1u = u$

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if $P_1 = (x_1, \dots, x_n)$, $P_2 = (y_1, \dots, y_n)$ points in $\mathbb{R}^n$

then the vector $v = \overrightarrow{P_1P_2} = (y_1 - x_1, y_2 - x_2, \dots, y_n - x_n)$

Ex: If $v = \overrightarrow{P_1P_2}$ where $P_1 = (2, -1, 4)$, $P_2 = (7, 5, -8)$

find v, ($v = (5, 6, -12)$)

Def: length (norm) (magnitude)

Let $u = (u_1, u_2, \dots, u_n)$ vector in $\mathbb{R}^n$

norm $u = \|u\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$ is real no.

$\|cu\| = |c| \|u\|$

Dot product

Let $u = (u_1, u_2, \dots, u_n)$, $v = (v_1, v_2, \dots, v_n)$ vectors in $\mathbb{R}^n$

the dot product or the inner product between u and v is

$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n = \|u\| \|v\| \cos\theta$

θ is the angle between u and v, $0 \le \theta \le \pi$

$\|u\| = (u \cdot u)^{\frac{1}{2}} = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$, $\|u\|^2 = u \cdot u$

$\cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$

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Ex: Let $u = (0,0,1)$, $v = (0,2,2)$, find $u.v$ $(u.v = 2)$

Ex: Let $u = (2,-1,1)$, $v = (1,1,2)$ ① find angle between u and v $\left(\theta = \frac{\pi}{3}\right)$.

② find $\|v - u\|$

③ Show that $u.v = \frac{1}{2}\left(\|u\|^2 + \|v\|^2 - \|v-u\|^2\right)$

Properties of dot product

If u, v and w are vectors in $\mathbb{R}^n$, $k \in \mathbb{R}$, then

① $u.u = \|u\|^2 \geq 0$ $(u.u = 0$ only if $u = 0)$

② $u.v = v.u$

③ $u.(v+w) = u.v + u.w$

④ $k.(u.v) = ku.v = u.kv$.

if $u.v > 0$, $\Leftrightarrow$ θ is acute

$u.v < 0$ $\Leftrightarrow$ θ is obtuse

$u.v = 0$ $\Leftrightarrow$ $\theta = \frac{\pi}{2}$ (u and v are perpendicular or orthogonal)

$u.v = \|u\|\|v\| \overset{\theta=0}{\Leftrightarrow}$ u and v are parallel in the same direction

$u.v = -\|u\|\|v\| \Leftrightarrow \theta = \pi$, u and v are parallel in the opposite direction

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Ex: Let $u = (2, k)$ $v = (3, 5)$ Find k such that ($|u \cdot v| = \|u\| \|v\|$)

① u and v are parallel → $k = 10/3$

② " " " orthogonal → $k = -6/5$ ($u \cdot v = 0$)

③ angle between u and v is $\frac{\pi}{4}$ → $k = \frac{1}{2}, -8$. ($u \cdot v = \frac{1}{\sqrt{2}} \|u\| \|v\|$)

Ex: $u = (-1, 2, 5)$ find k such that $\|ku\| = 4$

$\left(k = \pm \frac{4}{\sqrt{30}}\right)$

Ex: show that

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

(Hint: $(u+v)\cdot(u+v) + (u-v)\cdot(u-v) = 2(u\cdot u) + 2(v\cdot v)$)

HW

Ex: Show that

$$u \cdot v = \frac{1}{4}\|u+v\|^2 - \frac{1}{4}\|u-v\|^2$$

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Chapter 4

(4.1) Vector spaces

Def:
let V be set of objects called vectors
with two operations
$\oplus$: vector addition $(u \oplus v)$ u, and v are vectors in V
$\odot$: scalar multiplication $(c \odot u)$, $c \in \mathbb{R}$, $u, v \in V$.
If the following properties are satisfied, then
V is called vector space
(where u, v, w are vectors in V, c, k are constants)

(1) If u and v are vectors in V, then $u \oplus v \in V$
(2) $u \oplus v = v \oplus u$
(3) $u \oplus (v \oplus w) = (u \oplus v) + w$
(4) There is zero vector $0_v \in V$ such that $0_v \oplus u = u \oplus 0_v = u$
(5) There is negative of u $(-u) \in V$ such that $-u \oplus u = u \oplus -u = 0_v$
(6) If $c \in \mathbb{R}$, $u \in V$, then $c \odot u \in V$
(7) $c \odot (u \oplus v) = c \odot u \oplus c \odot v$
(8) $(c + k) \odot u = c \odot u \oplus k \odot u$
(9) $c \odot (k \odot u) = ck \odot u$
(10) $1 \odot u = u$.

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If $V = \mathbb{R}^n$, then the elements are of the form

$$u = (u_1, u_2, \dots, u_n)$$
$$v = (v_1, v_2, \dots, v_n)$$

If $V = M_{n,m}$, then the elements are of the form

$A = [a_{ij}]$ $n \times m$ matrix

If $V = P_n$ (polynomials of order n) then the elements are of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

i.e.

$V = \mathbb{R}^2$ the elements of the form

$u = (u_1, u_2)$ $u_1, u_2 \in \mathbb{R}$

$V = \mathbb{R}^3 \rightarrow u = (a, b, c) \in V$

$V = M_{2,2} \rightarrow A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in V$

$V = P_1 \rightarrow f(x) = ax + b \in V$

$V = P_2 \rightarrow f(x) = ax^2 + bx + c \in V$

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Examples

Ex1: $V = 0$

The zero vector space that contain only the zero vector

V is vector space satisfies all the conditions

Ex2: $V = \mathbb{R}^n$ $\quad u \oplus v = u + v$

$c \odot u = cu$

V is vector space under usual addition and usual scalar multiplication

Ex3: $V = M_{2,2} = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} / a,b,c,d \in \mathbb{R} \right\}$

$A \oplus B = A + B$

$c \odot A = cA$

then V satisfies the 10th conditions

so V is vector space.

Ex4: $V = P_n = \left\{ f(x) = a_n x^n + \cdots + a_1 x + a_0 / a_i \in \mathbb{R} \right\}$

$f(x) \oplus g(x) = f(x) + g(x)$

$c \odot f(x) = cf(x)$

V is vectorspace.

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Ex: Let $V = \mathbb{R}^2$

Define the operations

$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2 - 1, y_1 + y_2 + 2)$

$c \odot (x, y) = (2cx, c + y)$

Find $3 \odot ((2,1) \oplus (3,-2))$

$3 \odot ((2,1) \oplus (3,-2))$

$= 3 \odot (2+3-1, 1-2+2)$

$= 3 \odot (4, 1)$

$= (2(3)(4), 3+1) = (24, 4)$

Ex: Let $V = \mathbb{R}^2$

$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2 + 3, y_1 + y_2 - 1)$

$c \odot (x, y) = (cx, cy)$

Find 0_V

Let $0_V = (a, b)$, $u = (x_1, y_1)$

$u \oplus 0_V = u$

$(x_1, y_1) \oplus (a, b) = (x_1, y_1)$

$(x_1 + a + 3, y_1 + b - 1) = (x_1, y_1)$

$\rightarrow x_1 + a + 3 = x_1 \rightarrow a = -3 \Rightarrow 0_V = (-3, 1)$

and $y_1 + b - 1 = y_1 \rightarrow b = 1$

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Ex: let $V = R^2$

Define $(u_1, u_2) \oplus (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$

$c \odot (u_1, u_2) = (cu_1, 0)$

Is V vector space

No axiom 10 $\quad 1 \odot u = u$

Counter example

let $u = (2, 3)$

$1 \odot u = 1 \odot (2, 3) = (1(2), 0) = (2, 0) \neq (2, 3)$

So V is not vector space.

Ex: $V = R^3$ $\quad u \oplus v = u + v$, $\quad c \odot u = (cu_1, u_2, u_3)$ is V vectorspace?

No axiom 8 $\quad (c+k) \odot u \stackrel{?}{=} c \odot u \oplus k \odot u$

Counter ex let $c = 2$, $k = 3$, $u = (1, 1, 1)$

$(c+k) \odot u = 5 \odot (1, 1, 1) = (5, 1, 1)$

$c \odot u \oplus k \odot u = 2 \odot u \oplus 3 \odot u = (2, 1, 1) \oplus (3, 1, 1) = (5, 2, 2)$ $\neq$

So $(c+k) \odot u + c \odot u \oplus k \odot u \rightarrow V$ is not vector space

Ex: let $V = P_2$

$f \oplus g = f - g$ $\quad$ is V vector space?

$c \odot f = cf$

No let $f(x) = 3x^2 + 2x + 1$ $\quad g(x) = x^2 + 2x + 3$

$f \oplus g = f - g = 2x^2 - 2$

$g \oplus f = g - f = -2x^2 + 2$ $\neq$ So V is not vector space

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Theorem

If V be vector space, $k \in \mathbb{R}$, then

(a) $0 \odot u = 0_V$

(b) $k \odot 0_V = 0_V$

(c) $-1 \odot u = (-u)_V$

(d) If $k \odot u = 0_V$, then $k = 0$ or $u = 0_V$

Ex: $V = \mathbb{R}^2$

$(u_1, u_2) \oplus (v_1, v_2) = (u_1 + v_1 + 1, u_2 + v_2 + 1)$

$c \odot u = (cu_1, cu_2)$

$0_V = (-1, -1)$

$0 \odot (u_1, u_2) = (0, 0) \neq (-1, -1) = 0_V$

So V is not vector space.

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(4.2) Subspaces

Def:

A subset W of a vector space V is called a subspace of V if W itself a vector space under the addition and scalar multiplication defined on V.

Theorem:

If W is set of vectors in a vector space V, then W is a subspace of V if and only if

(a) If u and v are vectors in W, then $u \oplus v$ is in W

(b) If c is any scalar, $u \in W$, then $c \odot u \in W$

* Note

we will consider the vector space are under natural addition of vectors and natural scalar multiplication

i.e. $u \oplus v = u + v$

$c \odot u = cu$

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W is not subspace of one of the conditions does not satisfy

That is

give an example (called counter example)

$u \in W$, $v \in W$ and $u \oplus v \notin W$

OR example for the 2nd cond^n

$c \in \mathbb{R}$ $u \in W$

and $cu \notin W$.

if we have one of these examples then W is not subspace

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Examples

(1) $W = \{(x,y) \ / \ x \geq 0, y \geq 0\}$

(2) $W = \{A \in M_{n,n} \ / \ A \text{ is symmetric}\}$

(3) $W = \{A \in M_{2,2} \ / \ A \text{ is invertible}\}$

(4) $W = \{f \in P_2 \ / \ f(x) = a_2x^2 + a_1x + a_0, \ a_2 \neq 0\}$

(5) $W = \{\begin{bmatrix} a & a \\ b & b \end{bmatrix} \ / \ a, b \in \mathbb{R}\}$

(6) $W = \{(a, b, 1) \ / \ a, b \in \mathbb{R}\}$.

(7) $W = \{(a, a^2) \ / \ a \in \mathbb{R}\}$

(8) $W = \{(a, b, a-b) \ / \ a, b \in \mathbb{R}\}$

(9) $W = \{f = ax^2 + bx + c \ / \ a = -c\}$

(10) $W = \{X \in \mathbb{R}^n \ / \ AX = 0 \text{ for some } n \times n \text{ matrix } A\}$

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Examples

① $W = \{(x,y) \mid x \geq 0, y \geq 0\}$

is W subspace

if 1) $u \in W$, $v \in W$ then $u+v \in W$

$u = (x_1, y_1) \in W \to x_1 \geq 0, y_1 \geq 0$, $v = (x_2, y_2) \in W$ $x_2, y_2 \geq 0$

$u+v = (x_1+x_2, y_1+y_2)$ $x_1+x_2 \geq 0$ and $y_1+y_2 \geq 0$ ✓

$\to u+v \in W$

$c \in \mathbb{R}$ $u \in W$ $u = (x,y)$ $x, y \geq 0$

$cu = (cx, cy)$ for any $c \in \mathbb{R}$

is $cu \in W$. No counter ex.

Let $c = -2$, $u = (2,3) \in W$

$-2u = (-4, -6) \notin W$.

② $W = \{A \in M_{n,n} \mid A \text{ is symmetric}\}$

A is sym $\iff A = A^t$

Let $A, B \in W \to A = A^t$ and $B = B^t$

$(A+B)^t = A^t + B^t = A + B \to A+B$ is sym

$A, B \in W \to A+B \in W$

Let $c \in \mathbb{R}$, $A \in W$ $(A = A^t)$

$(cA)^t = cA^t = cA \to cA \in W$

$A \in W$

So W is subspace

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(3) $W = \{ A \in M_{2\times 2} \mid A \text{ is invertible} \}$

A is invertible $\iff \det A \neq 0$

W is not subspace

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \in W \quad (\det A \neq 0)$

$B = \begin{bmatrix} -2 & -1 \\ -1 & -3 \end{bmatrix} \in W \quad (\det B \neq 0)$

but $A+B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin W \quad (\det(A+B) = 0) \rightarrow W$ is not subspace

OR $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \in W \qquad 0 \in \mathbb{R}$

$0A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin W \rightarrow W$ is not subspace

(4) $W = \{ f \in P_2 \mid a_2 \neq 0 \}$

W is not subspace

ex. $f = 3x^2 - 2x + 1 \qquad g = -3x^2 + x - 2$

$f + g = -x - 1 \notin W$

(5) $W = \left\{ \begin{bmatrix} a & a \\ b & b \end{bmatrix} \mid a, b \in W \right\}$

Let $A = \begin{bmatrix} a_1 & a_1 \\ b_1 & b_1 \end{bmatrix} \in W, \quad B = \begin{bmatrix} a_2 & a_2 \\ b_2 & b_2 \end{bmatrix} \in W$

$A+B = \begin{bmatrix} a_1+a_2 & a_1+a_2 \\ b_1+b_2 & b_1+b_2 \end{bmatrix} \in W \rightarrow$

Let $c \in \mathbb{R} \quad A = \begin{bmatrix} a & a \\ b & b \end{bmatrix}$

$cA = \begin{bmatrix} ca & ca \\ cb & cb \end{bmatrix} \in W$

So W is subspace

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(6) $W = \{(a, b, 1) \mid a, b \in \mathbb{R}\}$

Let $u = (2, 3, 1)$, $v = (1, 1, 1)$ in W

$u + v = (3, 4, 2) \notin W$ so W is not subspace of $\mathbb{R}^3$

(7) $W = \{(a, a^2) \mid a \in \mathbb{R}\}$

$u = (2, 4) \in W$, $v = (3, 9) \in W$

$u + v = (5, 13) \notin W \rightarrow W$ is not subspace of $\mathbb{R}^2$

$\neq 25$

(8) $W = \{(a, b, a-b) \mid a, b \in \mathbb{R}\}$

Let $u = (a_1, b_1, a_1 - b_1)$, $v = (a_2, b_2, a_2 - b_2)$

$u + v = (a_1 + a_2, b_1 + b_2, a_1 + a_2 - b_1 + a_2 - b_2)$

$= (a_1 + a_2, b_1 + b_2, a_1 + a_2 - (b_1 + b_2)) \in W$

Let $c \in \mathbb{R}$, $u = (a, b, a-b) \in W$

$cu = (ca, cb, c(a-b)) = (ca, cb, ca - cb) \in W$

$\rightarrow W$ is subspace of $\mathbb{R}^3$

(9) $W = \{ax^2 + bx + c \mid a = -c\}$

$\rightarrow W = \{ax^2 + bx - a \mid a, b \in \mathbb{R}\}$

Let $f = a_1x^2 + b_1x - a_1$, $g = a_2x^2 + b_2x - a_2 \in W$

$f + g = (a_1 + a_2)x^2 + (b_1 + b_2)x - (a_1 + a_2) \in W$

$c \in \mathbb{R}$ $cf = ca_1x^2 + cb_1x - ca_1 \in W \rightarrow W$ subspace of P_2

(10) $W = \{X \in \mathbb{R}^n \mid AX = 0$, some $A\}$

Let $X \in W (AX = 0)$, $Y \in W (AY = 0)$

$A(X + Y) = AX + AY = 0 + 0 = 0 \rightarrow X + Y \in W$

Let $c \in \mathbb{R}$, $A \in W (AX = 0)$

$cAX = c0 = 0 \rightarrow cA \in W \rightarrow W$ is subspace of $\mathbb{R}^n$.

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(4.2) Linear Combination

Def:

A vector w is a linear combination of the vectors $v_1, v_2, \dots, v_r$ if it can be expressed as

$$w = c_1 v_1 + c_2 v_2 + \dots + c_r v_r \quad , c_i \in \mathbb{R}$$

Ex: $V = \mathbb{R}^2$

$(2,3) = 2(1,0) + 3(0,1)$ → $w = c_1 v_1 + c_2 v_2$

(w = $(2,3)$, v_1 = $(1,0)$, v_2 = $(0,1)$)

→ w is linear combination of v_1 and v_2

$v_1 = (3,1)$, $v_2 = (-1,2)$, $w = (3,8)$

$(3,8) = 2(3,1) + 3(-1,2)$ → $w = 2v_1 + 3v_2$

(w = $(3,8)$, v_1 = $(3,1)$, v_2 = $(-1,2)$)

→ w is linear combination of v_1, v_2

Ex: $\mathbb{R}^3$

$w_1 = (3,2,5)$, $v_1 = (3,2,5)$, $v_2 = (1,2,-1)$

$w_1 = (1)v_1 + 0v_2$ → w_1 is linear comb of v_1, v_2

$w_2 = (6,4,10)$ $w_2 = 2v_1 + 0v_2$

→ w_2 linear comb. of v_1, v_2.

Ex: $V = P_1$

$f_1 = 2+x$, $f_2 = -1+2x$, $f_3 = 5-x$

$2f_1 - f_2 + f_3 = 4+2x+1-2x+5-x = 10-x = f$

f is linear comb of $\{f_1, f_2, f_3\}$

$f = 2f_1 - f_2 + f_3$

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Ex: Consider $u=(1,2,-1)$, $v=(6,4,2)$ in $\mathbb{R}^3$

Show that $w=(9,2,7)$ is linear comb. of u and v and $w'=(4,1,8)$ is not linear comb. of u and v.

$w = c_1u + c_2v$ ⇝ linear system in c_1, c_2

- has soln unique or infinite → w is linear comb
- No soln → not linear comb.

$$(9,2,7) = c_1(1,2,-1) + c_2(6,4,2)$$

$$(9,2,7) = (c_1+6c_2,\ 2c_1+4c_2,\ -c_1+2c_2)$$

$$\begin{aligned} c_1+6c_2 &= 9 \\ 2c_1+4c_2 &= 2 \\ -c_1+2c_2 &= 7 \end{aligned} \longrightarrow \begin{bmatrix} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 & 9 \\ 0 & -8 & -16 \\ 0 & 8 & 16 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 9 \\ 0 & 1 & 2 \\ 0 & 8 & 16 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$c_1 = -3$, $c_2 = 2$ → $w = -3u + 2v$.

→ w is linear comb of $\{u, v\}$

$w' = (4,1,8)$

$$\begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & 1 \\ -1 & 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -7 \\ 0 & 8 & 12 \end{bmatrix} \xrightarrow{-2R_1+R_2} = \begin{bmatrix} 1 & 6 & 4 \\ 0 & 1 & 7/8 \\ 0 & 8 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 6 & 4 \\ 0 & 1 & 7/8 \\ 0 & 0 & 5 \end{bmatrix} \rightarrow 0 = 5 \text{ No soln}$$

→ w' is not linear comb of u, v.

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Ex: Express the vector $P = -9 - 7x - 15x^2$ as linear combination of $P_1 = 2 + x + 4x^2$, $P_2 = 1 - x + 3x^2$, $P_3 = 3 + 2x + 5x^2$

P is linear comb. of P_1, P_2, P_3

is there c_1, c_2, c_3 $\quad P = c_1P_1 + c_2P_2 + c_3P_3$

$$\begin{bmatrix} 2 & 1 & 3 & -9 \\ 1 & -1 & 2 & -7 \\ 4 & 3 & 5 & -15 \end{bmatrix} \xrightarrow{\text{use elementary row operations}} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$c_1 = -2, \quad c_2 = 1, \quad c_3 = -2$

$P = -2P_1 + P_2 - 3P_3$

$\rightarrow$ P is linear comb. of P_1, P_2, P_3

Ex: let $P_1 = 1 - x$, $\quad P_2 = 2 - 2x$

is $P = -2 + 2x$ linear combination of P_1, P_2

is there $c_1, c_2 \quad P = c_1P_1 + c_2P_2$

$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$(c_1, c_2) = (-2 - 2t, t)$

$t = 0 \quad (c_1, c_2) = (-2, 0)$

$t = 1 \quad (-4, 1)$

$P = -2P_1 + 0P_2$ or $= -4P_1 + P_2$.

$\rightarrow$ P is linear comb. of P_1, P_2.

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4.2 Span

Def:

$S = \{v_1, v_2, \dots, v_r\}$ set of vectors in a vector space V. if u is any vector in V and u is linear combination of the vectors in S (i.e. there are $c_1, \dots, c_r$, $u = c_1v_1 + c_2v_2 + \dots + c_rv_r$) then $u \in \text{Span } S$ and $V = \text{span } S$

i.e. $V = \text{span}\{v_1, \dots, v_r\}$

Ex1: Let $V = \mathbb{R}^3$

$S = \{(1,0,0), (0,1,0), (0,0,1)\}$

Determine whether S span $\mathbb{R}^3$.

Let $u \in \mathbb{R}^3 \rightarrow u = (a,b,c)$

$(a,b,c) = a(1,0,0) + b(0,1,0) + c(0,0,1)$

c_1 c_2 c_3

so $u \in \text{span } S \rightarrow \mathbb{R}^3 = \text{span } S$

or S span $\mathbb{R}^3$

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Ex 2: Determine whether $v_1 = (1, 1, 2)$ $v_2 = (1, 0, 1)$ and $v_3 = (2, 1, 3)$ span the vector space $\mathbb{R}^3$?.

Let $u = (a, b, c) \in \mathbb{R}^3$

is $u \in \text{span}\{v_1, v_2, v_3\}$

is u linear combination of v_1, v_2, v_3

is there c_1, c_2, c_3 $\quad u = c_1 v_1 + c_2 v_2 + c_3 v_3$

$$(a, b, c) = c_1(1, 1, 2) + c_2(1, 0, 1) + c_3(2, 1, 3)$$

Solve the linear system

$$\begin{bmatrix} 1 & 1 & 2 & a \\ 1 & 0 & 1 & b \\ 2 & 1 & 3 & c \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & b \\ 0 & 1 & 1 & a-b \\ 0 & 0 & 0 & c-a-b \end{bmatrix}$$

No Soln if $c-a-b \neq 0$

OR for square linear systems

$AX = B$ if $\det A \neq 0 \rightarrow$ there is No unique soln

$$\det \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix} = 0 \rightarrow \mathbb{R}^3 \neq \text{span}\{v_1, v_2, v_3\}.$$

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Ex3: Determine whether the given vectors span $\mathbb{R}^3$

(a) $v_1 = (2,2,2)$, $v_2 = (0,0,3)$, $v_3 = (0,1,1)$

(b) $v_1 = (2,-1,3)$, $v_2 = (4,1,2)$, $v_3 = (8,-1,8)$

(c) $v_1 = (3,1,4)$, $v_2 = (2,-3,5)$, $v_3 = (5,-2,9)$, $v_4 = (1,4,-1)$

(d) $v_1 = (1,2,6)$, $v_2 = (3,4,1)$, $v_3 = (4,3,1)$, $v_4 = (3,3,1)$

(a) $\det \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 1 \\ 2 & 3 & 1 \end{bmatrix} = -6 \neq 0$

So v_1, v_2, v_3 span $\mathbb{R}^3$

(b) $\det \begin{bmatrix} 2 & 4 & 8 \\ -1 & 1 & -1 \\ 3 & 2 & 8 \end{bmatrix} = 0$

So v_1, v_2, v_3 do not span $\mathbb{R}^3$.

(c) $\begin{bmatrix} 3 & 2 & 5 & 1 & a \\ 1 & -3 & -2 & 4 & b \\ 4 & 5 & 9 & -1 & c \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -3 & -2 & 4 & * \\ 0 & 11 & 11 & -11 & * \\ 0 & 0 & 0 & 0 & * \end{bmatrix}$

No Solns

So v_1, v_2, v_3, v_4 do not span $\mathbb{R}^3$

(d) $\begin{bmatrix} 1 & 3 & 4 & 3 & * \\ 2 & 4 & 3 & 3 & * \\ 6 & 1 & 1 & 1 & * \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 3 & 4 & 3 & * \\ 0 & 1 & 5/2 & 3/2 & * \\ 0 & 11 & 8 & 8 & * \end{bmatrix}$ ← Lots

→ ~~Unit~~ ther is Soln (infinite)

→ v_1, v_2, v_3, v_4 span $\mathbb{R}^3$.

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Ex 4 Does $f_1 = x + 2x^2$, $f_2 = 4 + 4x + 6x^2$ are in span $\{\underbrace{-1 + x + 2x^2}_{g_1}, \underbrace{2 + 2x - 3x^2}_{g_2}, \underbrace{4 - x^2}_{g_3}\}$

$f_1, f_2 \in \text{span } S$

$\Leftrightarrow$ f_1, f_2 are linear comb of g_1, g_2, g_3

$[g_1\ g_2\ g_3\ f_1]$ and $[g_1\ g_2\ g_3\ f_2]$ has soln ?

$$\left[\begin{array}{ccc:c:c} -1 & 2 & 4 & 0 & 4 \\ 1 & 2 & 0 & 1 & 4 \\ 2 & 3 & -1 & 2 & 6 \end{array}\right] \quad \begin{matrix} & & & f_1 & f_2 \end{matrix}$$

$$\rightsquigarrow \left[\begin{array}{ccc:c:c} 1 & 0 & -2 & \frac{1}{2} & 0 \\ 0 & 1 & 1 & \frac{1}{4} & 2 \\ 0 & 0 & 0 & \frac{1}{4} & 0 \end{array}\right]$$

No soln

infinite soln

So $f_1 \notin \text{span } S$

$f_2 \in \text{span } S$

Ex: Is $u = (-3, 6, -9) \in \text{span} \underbrace{\{(2, -4, 6), (1, -2, 2)\}}_{S}$

$$\left[\begin{array}{cc:c} 2 & 1 & -3 \\ -4 & -2 & 6 \\ 6 & 2 & -9 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc:c} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \rightarrow u \in \text{span } S$$

$(c_1, c_2) = (-5, 1)$ or $(-3, 0)$ or $(-1, -1)$

$c_1 = -3 - 2c_2$

$u = -5u_1 + u_2$ or $u = -3u_1 + 0u_2$ or $u = -u_1 - u_2$

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Ex5 Find all values of a so that $(a^2, a, 1) \in \text{span}\{(1,2,3),(1,1,1),(a,1,2)\}$

$$\begin{bmatrix} 1 & 1 & a & a^2 \\ 2 & 1 & 1 & a \\ 3 & 1 & 2 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & a-a^2 \\ 0 & 1 & -1 & 2a^2-a \\ 0 & 0 & 0 & a^2-2a+1 \end{bmatrix}$$

there is sol^n if $a^2-2a+1=0$

$\rightarrow (a-1)(a-1)=0 \rightarrow a=1$.

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(4.3) Linear independence

Definition:

If $S = \{v_1, v_2, \ldots, v_r\}$ is set of vectors in a vector space V, then

S is linearly independent iff

$c_1v_1 + c_2v_2 + \cdots + c_rv_r = 0_V$ has only trivial solution (i.e $c_1 = c_2 = \cdots = c_r = 0$)

If the solution is infinite number, then S is linearly dependent.

To test $S = \{v_1, v_2, \ldots, v_r\}$ if linearly indep. or linearly dependent

Solve $c_1v_1 + c_2v_2 + \cdots + c_rv_r = 0_V \rightsquigarrow [v_1\ v_2 \ldots v_r\ 0]$ solve

- one solution (only trivial) $c_1 = c_2 = \cdots = c_r = 0$ → S is lin. indep
- infinite solution, at least one $c_i \neq 0$ → S is lin. dep.

* If the matrix $[v_1 \ldots v_r]$ is square use

$\det[v_1 \ldots v_r]$ $\neq 0 \rightarrow S$ lin. indep.

$= 0 \rightarrow S$ lin. dep.

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Ex 1: Determine whether the vectors

$v_1 = (1,-2,3)$, $v_2 = (5,6,-1)$, $v_3 = (3,2,1)$

if linearly dep. or in dep.

$$\det \begin{bmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{bmatrix} = 1(8) + 2(8) + 3(-8) = 0$$

$\det A \neq 0$
$\leftrightarrow AX = 0 \rightarrow X = 0$

→ nontrivial soln → v_1, v_2, v_3 are linearly dependant

$c_1v_1 + c_2v_2 + c_3v_3 = 0_v$

Find relation between v_1, v_2, v_3

$\det A \neq 0 \rightarrow$ in dep.
$\det A = 0 \rightarrow$ dep

$$\begin{bmatrix} 1 & 5 & 3 & 0 \\ -2 & 6 & 2 & 0 \\ 3 & -1 & 1 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$(c_1, c_2, c_3) = (\frac{-1}{2}t, \frac{-1}{2}t, t)$ $t \in \mathbb{R}$

Let $t = 2 \rightarrow (c_1, c_2, c_3) = (-1, -1, 2)$

$-v_1 - v_2 + 2v_3 = (0,0,0)$ or $\rightarrow 2v_3 = v_1 + v_2$

Ex 2: $v_1 = (1,2,2,-1)$, $v_2 = (4,9,9,-4)$, $v_3 = (5,8,9,-5)$

linearly dep or in dep.

solve $c_1v_1 + c_2v_2 + c_3v_3 = 0_v = (0,0,0,0)$

Solve $\begin{bmatrix} 1 & 4 & 5 & 0 \\ 2 & 9 & 8 & 0 \\ 2 & 9 & 9 & 0 \\ -1 & -4 & -5 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$c_1 = c_2 = c_3 = 0$ only trivial

→ v_1, v_2, v_3 linearly in dep.

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Ex 3 : $P_1(x) = 1-x$, $P_2(x) = 5+3x-2x^2$

$P_3(x) = 1+8x-x^2$ are P_1, P_2, P_3 linearly dep ?

$$\det\begin{bmatrix} 1 & 5 & 1 \\ -1 & 3 & 3 \\ 0 & -2 & -1 \end{bmatrix} = 1(3) + 1(-3) = 0$$

→ P_1, P_2, P_3 linearly dependent.

Find this dependence. → solve $c_1P_1 + c_2P_2 + c_3P_3 = 0$

$$\begin{bmatrix} 1 & 5 & 1 & 0 \\ -1 & 3 & 3 & 0 \\ 0 & -2 & -1 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -3/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$(c_1, c_2, c_3) = \left(\frac{3}{2}t, \frac{-1}{2}t, t\right)$ $t \in \mathbb{R}$

$(c_1, c_2, c_3) = (3, -1, 2)$

$3P_1 - P_2 + 2P_3 = 0$ or $P_2 = 3P_1 + 2P_3$

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Theorem:

(a) any set of vectors contain zero vector is linearly dependent.

(b) $S=\{v\}$ S has one vector is linearly indep iff $v \neq 0_v$.

(c) $S=\{v_1, v_2\}$ S is linearly indep. iff neither one vector is multiple of the other vector. ($v_i \neq cv_2$)

Theorem:

Let $S=\{v_1, v_2, \cdots, v_r\}$ be set of vectors in $\mathbb{R}^n$

If $r>n$, then S is linearly dependent.

Proof: the homogeneous system

$c_1 v_1 + c_2 v_2 + \cdots + c_r v_r = 0_v$

$[v_1\ v_2 \cdots v_r | 0_v]$ this matrix has

n rows and $r+1$ columns

no. of rows $<$ no. of columns

$\rightarrow$ homog. system has infinite no. of sol^{n}s

$\rightarrow$ non trivial sol^n

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Ex: Which of the following set of vectors are linearly dep.

(a) $(4,-1,2)$, $(-4,10,2)$

$c_1v_1 + c_2v_2 = 0_v \rightsquigarrow \begin{bmatrix} 4 & -4 & 0 \\ -1 & 10 & 0 \\ 2 & 2 & 0 \end{bmatrix}$ solve $\rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

→ $c_1 = c_2 = 0$ → only trivial sol^n

→ linearly indep.

(b) $(-3,0,4)$, $(5,-1,2)$, $(1,1,3)$

$\det \begin{bmatrix} -3 & 5 & 1 \\ 0 & -1 & 1 \\ 4 & 2 & 3 \end{bmatrix} = -3(-5) + 4(6) = 39 \neq 0 \iff c_1 = c_2 = c_3 = 0$

→ linearly indep.

(c) $(-2,0,1)$, $(3,2,5)$, $(6,-1,1)$, $(7,0,-2)$

$\{v_1, v_2, v_3, v_4\} \subseteq \mathbb{R}^3$ $\quad 4 > 3$

→ they are linearly dep.

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(4.4) Coordinates and Basis

Definition (Basis)

If V is vector space and $S=\{v_1, v_2, \dots, v_n\}$ is finite set of vectors in V, then S is called a basis for V if the following two conditions satisfied

(a) S is linearly independent $\begin{cases} c_1v_1+c_2v_2+\dots+c_nv_n=0 \\ \rightarrow c_1=c_2=\dots=c_n=0 \\ [v_1\ v_2\ \dots\ v_n\ 0] \\ \text{or } \det[v_1 \dots v_n]\neq 0 \end{cases}$

(b) S spans V $\rightarrow \{v\in V$ v is linear comb. of $S \rightarrow v=c_1v_1+c_2v_2+\dots+c_nv_n\}$

Ex1: Let $V=R^3$ and

$S=\{(1,0,0),(0,1,0),(0,0,1)\}$

show that S is basis for R^3

(a) S is linearly indep.

$\det\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=1\neq 0 \rightarrow$ linearly indep.

(b) S spans $V=R^3$

if $v=(a,b,c)\in R^3$ $\quad c_1v_1+c_2v_2+c_3v_3=v$

$$\left[\begin{array}{ccc|c}1&0&0&a\\0&1&0&b\\0&0&1&c\end{array}\right] \rightarrow \begin{matrix}c_1=a\\c_2=b\\c_3=c\end{matrix} \rightarrow S \text{ spans } V \rightarrow S \text{ basis for } V$$

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Ex: Let $V = \mathbb{R}^2$

$S = \{(1,2), (2,5)\}$

Is S form basis for $\mathbb{R}^2$?

(a) $\det\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = 1 \neq 0 \rightarrow$ linearly indep.

(b) let $v = (a,b) \in \mathbb{R}^2$ are there c_1, c_2 $v = c_1 v_1 + c_2 v_2$

$$\begin{bmatrix} 1 & 2 & a \\ 2 & 5 & b \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 5a-2b \\ 0 & 1 & b-2a \end{bmatrix} \rightarrow \begin{matrix} c_1 = 5a-2b \\ c_2 = b-2a \end{matrix}$$

$\rightarrow S$ spans V

$\rightarrow S$ is basis for $\mathbb{R}^2$.

Ex: $V = P_1$

$S = \{2x+1, 6x+3\}$ is S basis for P_1

(a) is S linearly indep. $c_1 v_1 + c_2 v_2 = 0 \overset{?}{\rightarrow} c_1 = c_2 = 0$

$\det\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix} = 0 \rightarrow S$ is linearly dependent

$\rightarrow S$ is not basis for P_1.

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Definition: (Coordinate)

If $S = \{v_1, v_2, \cdots, v_n\}$ is basis for V

and $v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$

then $c_1, \cdots, c_n$ are called coordinates of v relative to S and denoted by

$(v)_S = (c_1, c_2, \cdots, c_n)$. is the coordinate vector

Ex: Let $V = \mathbb{R}^3$

and $S = \{(1,2,1), (2,9,0), (3,3,4)\}$ form a basis for $\mathbb{R}^3$. Find the coordinate vector of $v = (5, -1, 9)$ relative to S

Solution:

$v = c_1 v_1 + c_2 v_2 + c_3 v_3$

$$\begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 9 & 3 & -1 \\ 1 & 0 & 4 & 9 \end{bmatrix} \xrightarrow{\text{solve}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$\Rightarrow (v)_S = (c_1, c_2, c_3) = (1, -1, 2)$

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Ex: Find the coordinate vector of $p = 2 - x + x^2$ relative to the basis

$$S = \{ \underbrace{1+x}_{p_1}, \underbrace{1+x^2}_{p_2}, \underbrace{x+x^2}_{p_3} \}$$

$$p = c_1 p_1 + c_2 p_2 + c_3 p_3$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\rightarrow (p)_S = (c_1, c_2, c_3) = (0, 2, -1)$$

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(4.5) Dimension

Definition:

The dimension of a vector space V is the number of vectors in any basis for V and denoted by $\dim V$.

Theorem 1:

All bases for a finite-dimensional vector space have the same number of vectors.

Theorem 2:

Let V be vector space and $\{v_1, v_2, \cdots, v_n\}$ form basis for V

(a) If a set has more than n vectors, then it is linearly dependent

(b) If a set has fewer than n vectors, then it is does not span V.

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So if V is vector space and

$S = \{v_1, v_2, \dots, v_n\}$ is basis for V

i.e. $\dim V = n$, then

any set has vectors more than or less than n is not basis

Theorem:

If V is vector space $\dim V = n$

$B = \{u_1, u_2, \dots, u_n\}$, then

if B is linearly independent or spans V, then

B is basis for V

→ $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0_V$
→ $c_1 = c_2 = \dots = c_n = 0$
($AX = 0 \rightarrow X = 0$)
$\Leftrightarrow \det A \neq 0$

that is if $\det[u_1\ u_2 \dots u_n] \neq 0$ → linearly indep.
→ basis

if $\det[u_1\ u_2 \dots u_n] = 0$ → linearly dep
→ not basis

Ex: $V = \mathbb{R}^2$, $S = \{(1,0), (0,1)\}$ is basis → $\dim \mathbb{R}^2 = 2$

Ex: $V = \mathbb{R}^3$, $S = \{(1,0,0), (0,1,0), (0,0,1)\}$ is basis

→ $\dim \mathbb{R}^3 = 3$

Ex: $V = P_1$ $S = \{1, x\}$ is basis

→ $\dim P_1 = 2$

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Ex: let $V = P_2$, $S = \{1, x, x^2\}$ is basis

then $\dim P_2 = 3$

Ex: let $V = M_{2,2}$ $S = \left\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$

is basis then $\dim M_{2,2} = 4$.

So in general

$\dim \mathbb{R}^n = n$

$\dim P_n = n+1$

$\dim M_{n,m} = n \times m$.

Ex: let $V = P_2$

Determine which of the following set of vectors form basis for P_2

- $S_1 = \{2 - t + 4t^2, 3 + 6t + 2t^2, 2 + 10t - 4t^2\}$

$S_2 = \{2 - t + 3t^2, 5 + 2t + 3t^2\}$

$S_3 = \{2 + t + 2t^2, 1 - 2t, 2 + 3t^2, 1 + t + t^2\}$

$S_4 = \{2 + t^2, 1 + 2t - t^2, 1 - 2t + 2t^2\}$

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we know that $\dim P_2 = 3$

So directly S_2 and S_3 are not bases for P_2

since number of vectors $\neq 3$

$S_1 \xrightarrow[\text{and}]{\text{no. of vectors} = \dim P_2} \det \begin{bmatrix} 2 & 3 & 2 \\ -1 & 6 & 10 \\ 4 & 2 & -4 \end{bmatrix} = -32 \neq 0$

→ S_1 linearly independent

→ S_1 is basis for V

$S_4 \rightarrow \det \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & -2 \\ 1 & -1 & 2 \end{bmatrix} = 0$

→ S_4 linearly dependent

→ S_4 is not basis for V

Ex: Determine whether

$S = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \right\}$

form basis for $M_{2,2}$ or not.

1) no. of vectors in $S = \dim M_{2,2} = 4$

2) $\det \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} = 2 \neq 0$

A_1 A_2 A_3 A_4

→ S is basis

$\begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ row by row

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Basis and dimension for solution space (null space)

Given the homogeneous system $AX = 0$

Solve the homog. system

if the solution has only the trivial solution, then No basis and dimension = 0

if the solution is infinite number of solutions then the basis is the vectors form from the coefficients of $t, s, r, \ldots$ and the dimension is the number of these vectors.

Ex: Find basis and dimension for the solution space

a) $-3x - z = 0$
$2x + y = 0$
$-x + y - z = 0$

$\rightsquigarrow \begin{bmatrix} -3 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & -1 & 0 \end{bmatrix}$ Solve $\rightsquigarrow \begin{bmatrix} 1 & 0 & 1/3 & 0 \\ 0 & 1 & -2/3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\rightarrow (x, y, z) = \left(\frac{-1}{3}t, \frac{2}{3}t, t\right)$, $t \in \mathbb{R}$

$= \left(\frac{-1}{3}, \frac{2}{3}, 1\right)t$

$\rightarrow \left\{\left(\frac{-1}{3}, \frac{2}{3}, 1\right)\right\}$ is basis for solution space

and dimension = 1

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b) $x+2y+2z+5w=0$

$2x+4y+3z+7w=0$

$x+2y+z+2w=0$

$3x+6y+4z+9w=0$

$$\begin{bmatrix} 1 & 2 & 2 & 5 & 0 \\ 2 & 4 & 3 & 7 & 0 \\ 1 & 2 & 1 & 2 & 0 \\ 3 & 6 & 4 & 9 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$(x, y, z, w) = (s-2t, t, -3s, s) \quad s, t \in \mathbb{R}$

$$= \begin{bmatrix} s-2t \\ t \\ -3s \\ s \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} s \\ 0 \\ -3s \\ s \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \\ -3 \\ 1 \end{bmatrix} s$$

→ basis $= \{(-2, 1, 0, 0), (1, 0, -3, 1)\}$ dim = 2

c) $2x+3y+2z=0$

$-x+6y+10z=0$

$4x+2y-4z=0$

$$\begin{bmatrix} 2 & 3 & 2 & 0 \\ -1 & 6 & 10 & 0 \\ 4 & 2 & -4 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Solution $(x, y, z) = (0, 0, 0)$

→ No basis

dim = 0

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[4.7] Row space, Column space and Nullspace

If $A = [a_{ij}]$ is $m \times n$ matrix

Row space of A = span $\{R_1, R_2, \cdots, R_m\}$

that is $c_1R_1 + c_2R_2 + \cdots + c_mR_m \in$ Row space of A

Column space of A = span $\{c_1, c_2, \cdots, c_n\}$

i.e $k_1c_1 + k_2c_2 + \cdots + k_nc_n \in$ column space of A.

* How to find basis for rowspace of A

① Any Basis

Take A $\xrightarrow{\text{by elementary row operations}}$ $\begin{bmatrix} R_1' \\ R_2' \\ \vdots \\ R_j' \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

R.E matrix
(row echelon matrix)

Basis for row space of A = non zero rows in the R.E matrix
$= \{R_1', \cdots, R_j'\}$

② Basis for rowspace of A subset of rows of A.

Take A^t $\rightsquigarrow$ $[R.E]$ matrix

Columns that contain leading 1 determine which rows in A form basis.

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* How to find basis for column space of A

basis for column space of A ≡ basis for row space of A^t. so repeat previous steps for A^t.

That is (1.) for any basis (not necessarily subset of columns of A

Basis for column space of A

≡ Basis for row space of A^t

→ A^t ⇝ [R.E matrix]

nonzero rows = basis for row space of A^t

write them as column they will be basis for column space of A.

(2) Basis for column space of A (subset of A columns)

≡ " " row " = A^t (" = A^t rows)

Take $(A^t)^t = A$ ⇝ [R.E]

Column that contain leading 1's determine which rows in A^t are basis for rowspace of A^t

≡ which columns of A for basis for column space of A.

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Review

Basis for row space of A

① Any basis

$A \rightsquigarrow$ R.E matrix → non zero rows are basis for row space of A

② Subset of rows of A

$A^t \rightsquigarrow$ R.E as example $\begin{bmatrix} 1 & 2 & 3 & 2 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Leading 1 in c_1, c_3, c_4

→ Basis for row space = $\{R_1, R_3, R_4\}$ from A

Basis for column space of A

① Any basis

$A^t \rightsquigarrow$ R.E matrix

non zero row (writing them in columns) are basis for column space of A

② Subset of column of A

$A \rightsquigarrow$ R.E matrix as ex $\begin{bmatrix} 1 & -2 & 9 & 0 & 9 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

→ $\{c_1, c_2, c_5\}$ from A

basis for column space of A subset of A columns.

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Ex: Let $A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$

Find Basis for row space of A consisting entirely of row vectors of A.

$A^t = \begin{bmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 6 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{bmatrix}$ ⇝ $\begin{bmatrix} \textcircled{1} & 2 & 0 & 2 \\ 0 & \textcircled{1} & -5 & -10 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

c_{l_1} c_{l_2} c_{l_4}

→ $\{R_1, R_2, R_4\}$

$= \{[1\ -2\ 0\ 0\ 3], [2\ -5\ -3\ -2\ 6], [2\ 6\ 18\ 8\ 6]\}$

Basis for row space of A.

Ex: Find Bases for row space and column space of $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$

$A \equiv \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 1 & -19 \end{bmatrix} \equiv \begin{bmatrix} \textcircled{1} & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$

$\{[\textcircled{1}, -1, 3], [0, 1, -19]\}$ basis for row space of A

$\left\{\begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ -6 \end{bmatrix}\right\}$ " " column space of A

OR $A^t = \begin{bmatrix} 1 & 5 & 7 \\ -1 & -4 & -6 \\ 3 & -4 & 2 \end{bmatrix} \equiv \begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

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$\Rightarrow \left\{ \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ basis for column space of A

$\left\{ [1 \ -1 \ 3], [5 \ -4 \ -4] \right\}$ basis for rowspace of A

Ex: Find subset of the vectors

$u_1 = (1, -2, 0, 3)$, $u_2 = (2, -5, -3, 6)$

$u_3 = (0, 1, 3, 0)$, $u_4 = (2, -1, 4, 7)$, $u_5 = (5, -8, 1, 2)$

that form basis for the space spanned by these vectors.

$$A = \begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & -5 & -3 & 6 \\ 0 & 1 & 3 & 0 \\ 2 & -1 & 4 & 7 \\ 5 & -8 & 1 & 2 \end{bmatrix}$$

subset $\rightarrow$ take A^t

$$\begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & 7 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} \textcircled{1} & 0 & 2 & 0 & 1 \\ 0 & \textcircled{1} & -1 & 0 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rightarrow \{R_1, R_2, R_4\}$

$\rightarrow \{u_1, u_2, u_4\}$ basis for span $\{u_1, u_2, u_3, u_4, u_5\}$.

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Basis for null space of $A \equiv$ Basis for solution space $AX = 0$

Ex: Find a basis for the null space of

$$A = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix}$$

solution:

Solve $AX = 0$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 0 \end{bmatrix}$$

$$\rightsquigarrow \begin{bmatrix} \textcircled{1} & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -3x_2 - 4x_4 - 2x_5 \\ x_2 \\ -2x_4 \\ x_4 \\ x_5 \\ 0 \end{bmatrix}$$

$\rightarrow$ Basis for Null space $= \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

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(4.8) Rank and Nullity

Def (Rank)

If A is any $m \times n$ matrix, then the row space and column space have the same dimension. This dimension is called rank A

(Rank A = dim of row space = dim of column space)

$[A]_{m \times n} \rightsquigarrow [R.E]$ → number of nonzero rows = rank A.

Def (Nullity)

The dimension of nullspace of A ($\equiv$ basis for solution space $AX = 0$) is called nullity of A

Theorem:

If A is any matrix, then rank A = rank A^t

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Theorem:

If A is $m \times n$ matrix, then

$$\text{rank } A + \text{nullity } A = n$$

- rank A: number of leading variables in R.E of A
- nullity A: number of free variables in R.E of A
- n: number of columns of R.E of A

Ex: Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & 1 \\ 1 & 6 & -1 \end{bmatrix}$ Find rank A, nullity A

$$A \rightsquigarrow \begin{bmatrix} 1 & 0 & 4/7 \\ 0 & 1 & -3/7 \\ 0 & 0 & 0 \end{bmatrix}$$

rank $A = 2$

nullity $A = 1$

Ex: $A = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$ Find rank A, nullity A

$$A \rightsquigarrow \begin{bmatrix} 1 & 0 & -4 & -28 & -37 & 13 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

rank $A = 2$, nullity $A = 4$

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Ex: $A = \begin{bmatrix} 1 & -1 & 0 \\ 3 & 2 & 1 \\ 1 & 3 & 4 \end{bmatrix}$ Find rank A, nullity A

$A \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ rank $A = 3$
nullity $A = 0$

Ex: Let A be 5x7 matrix nullity $A = 3$
Find rank A.

nullity A + rank A = 7 (number of columns)

→ rank A = 4

Theorem:
The following statements are equivalent
For any nxn matrix A

① $\det A \neq 0$
② A^{-1} exist (A is invertible)
③ $AX = 0 \rightarrow X = 0$ (Solution of homog. system only trivial soln)
④ $AX = B \rightarrow X = A^{-1}B$ (Soln of linear system is unique)
⑤ rank A = n

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Ex: If A is 3×3 matrix and $\det A=-2$ find $\operatorname{rank} A$.

$\det A \neq 0 \rightarrow \operatorname{rank} A = 3$.

Ex: $A=\begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & -3 \\ 1 & 3 & 3 \end{bmatrix}$ find $\operatorname{rank} A$

$\det A = 0 \rightarrow \operatorname{rank} A < 3$ $\quad \operatorname{rank} A = 1$ or 2

$A \rightsquigarrow \begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \operatorname{rank} A = 2$.

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Chapter 5

Eigenvalues and eigenvectors

Def:

If A is an $n \times n$ matrix, then a nonzero vector $X \in \mathbb{R}^n$ is called an eigenvector of A if

$AX = \lambda X$ for some scalar λ

λ is called an eigenvalue of A

X is called an eigenvector corresponding to λ.

* How to find eigenvalues and eigenvectors

$AX = \lambda X \iff \lambda X - AX = 0 \iff (\lambda I - A)X = 0$

$(\lambda I - A)X = 0$ is homog system and the sol^n $X \neq 0$

so $\det(\lambda I - A) = 0$

by solving this equation we find the eigenvalues and for each eigenvalue

Solve the homog. system $(\lambda I - A)X = 0$ to find the solution which is the eigenvector corresponding to λ.

$p(\lambda) = \det(\lambda I - A)$ is called the charcteristic polynomial

and $\det(\lambda I - A) = 0$ is called the charcteristic equation

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Ex-1

Let $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$

Find the characteristic equation, the eigenvalues and their corresponding eigenvectors

$\det(\lambda I - A) = 0 \iff \det\left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}\right) = 0$

$\det \begin{bmatrix} \lambda - 1 & -1 \\ 2 & \lambda - 4 \end{bmatrix} = 0 \iff (\lambda - 1)(\lambda - 4) + 2 = 0$

$\lambda^2 - 5\lambda + 6 = 0$ is the charcteristic equation.

$(\lambda - 2)(\lambda - 3) = 0 \rightarrow$ eigenvalues $\lambda = 2, \lambda = 3$

eigenvector for $\lambda = 2$

Solve $(2I - A)X = 0 \iff \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}}_{2I - A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{0}$

$\begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow X = \begin{bmatrix} t \\ t \end{bmatrix}, t \in \mathbb{R}$

is the eigenvector corresponding to $\lambda = 2$

eigenvector for $\lambda = 3$

Solve $(3I - A)X = 0$

$\begin{bmatrix} 2 & -1 & 0 \\ 2 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow X = \begin{bmatrix} \frac{1}{2}t \\ t \end{bmatrix}, t \in \mathbb{R}$

is the eigenvector corresponding to $\lambda = 3$

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Ex 2 $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$

a) Find charcteristic polynomial

b) Find eigenvalues.

c) Find one eigenvector for any eigenvalue

a) $p(\lambda) = \det(\lambda I - A) = \det \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix}$

$= \lambda(\lambda^2 - 8\lambda + 17) + 1(0 - 4) = \lambda^3 - 8\lambda^2 + 17\lambda - 4$. The charact poly.

$= (\lambda - 4)(\lambda^2 - 4\lambda + 1)$

$\lambda = 4, \quad \lambda = \frac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3}$

$\begin{cases} ax^2 + bx + c \\ x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{cases}$

b) $\lambda = 4, \quad \lambda = 2 + \sqrt{3}, \quad \lambda = 2 - \sqrt{3}$.
are the eigenvalues.

c) eigenvector for $\lambda = 4$

Solve $(4I - A)X = 0 \rightsquigarrow \begin{bmatrix} 4 & -1 & 0 & 0 \\ 0 & 4 & -1 & 0 \\ -4 & 17 & -4 & 0 \end{bmatrix}$

$\rightsquigarrow \begin{bmatrix} 1 & 0 & -1/16 & 0 \\ 0 & 1 & -1/4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow X = \begin{bmatrix} 1/16\, t \\ 1/4\, t \\ t \end{bmatrix}$

is the eigenvector corresponding to $\lambda = 4$

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If A is diagonal matrix, upper triangular matrix or lower triangular matrix, then the eigenvalues of A are the entries of the diagonal.

Ex 3 $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 \\ 0 & 7 & 5 & 0 \\ 2 & 0 & -1 & 1/2 \end{bmatrix}$ Find eigenvalues

$\det(\lambda I - A) = (\lambda - 2)(\lambda + 1)(\lambda - 5)(\lambda - 1/2)$

$\Rightarrow$ eigenvalues $\lambda = 2, \lambda = -1, \lambda = 5, \lambda = \frac{1}{2}$.

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Chapter 8

(8.1) Linear transformation

Definition:

If $T: V \longrightarrow W$ is a function from a vector space V to a vector space W, then T is called linear transformation if

① $T(u+v) = T(u) + T(v)$, $u, v \in V$

② $T(ku) = kT(u)$ $\quad k \in \mathbb{R}$, $u \in V$

Theorem

If $T: V \longrightarrow W$ is linear transformation, then

$p \rightarrow q$ (if p then q)

$\sim q \rightarrow \sim p$ (if not q then not p)

(a) $T(0_V) = 0_W$

(b) $T(u-v) = T(u) - T(v)$.

Ex1: Zero Transformation $T(v) = 0_W$ show that T is linear transf.

$T: V \longrightarrow W$

let $u, v \in V$

(1) $T(u+v) = 0_W = 0_W + 0_W = T(u) + T(v)$ ✓

(2) $T(ku) = 0_W = k0_W = kT(u)$ ✓

then T is linear transformation

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Ex2: Identity operator

$T: V \to V \quad T(v) = v$. Show that T is linear

(1) $T(v_1 + v_2) = v_1 + v_2 = T(v_1) + T(v_2)$. ✓

(2) $T(kv) = kv = kT(v)$ ✓

So T is linear transformation

Ex3: $T: M_{2,2} \to M_{2,2}$

$T(A) = A^t$. Is T linear transf ?

(1) $T(A_1 + A_2) = (A_1 + A_2)^t = A_1^t + A_2^t = T(A_1) + T(A_2)$ ✓

(2) $T(cA) = (cA)^t = cA^t = cT(A)$ ✓

T is linear transf.

Ex4: $T: M_{2,2} \to \mathbb{R}$

$T(A) = \det A$ is T linear transf.

(1) $T(A_1 + A_2) = \det(A_1 + A_2) \neq \det A_1 + \det A_2$

$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A_2 \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ $\quad A_1 + A_2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

$T(A_1 + A_2) = \det(A_1 + A_2) = 9$

$T(A_1) + T(A_2) = 1 + 4 = 5$ $\Big\} \neq \to$ T is not linear transf

OR but $T(2A_1) = T\left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right) = \det\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 4$

$2T(A_1) = 2 \det A_1 = 2$ $\Big\} \neq \to$ T is not linear transf

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Ex5: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^2$

$T(x,y,z) = (x,z)$ Is T linear transf.

$u_1 = (x_1, y_1, z_1)$, $u_2 = (x_2, y_2, z_2)$

$T(u_1+u_2) = T((x_1+x_2, y_1+y_2, z_1+z_2)) = (x_1+x_2, z_1+z_2)$ =

$T(u_1) + T(u_2) = (x_1, z_1) + (x_2, z_2) = (x_1+x_2, z_1+z_2)$

1st condition ✓

$T(ku) = T(kx_1, ky_1, kz_1) = (kx_1, kz_1)$ =

$kT(u) = k\,T(x_1, y_1, z_1) = k(x_1, z_1) = (kx_1, kz_1)$

2nd condition ✓

So T is linear transf.

Ex6: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$T(x,y,z) = (x+1, 2y, z)$

Give Counter example to show that T is not linear transformation.

$u = (1,2,3)$, $v = (2,2,2)$, $u+v = (3,4,5)$

$T(u) + T(v) = (2,4,3) + (3,4,2) = (5,8,5)$ ≠

$T(u+v) = T(3,4,5) = (4,8,5)$

OR $u = (1,2,3)$ $k = 2$, $2u = (2,4,6)$

$T(2u) = T(2,4,6) = (3,8,6)$ ≠

$2T(u) = 2(2,4,3) = (4,8,6)$

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Ex 7: $T: P_2 \longrightarrow P_1$

$T(ax^2+bx+c) = cx + a-2$

Give counter example to show that T is not linear transformation.

Counter example:

$f(x) = 2x^2+3x+1$, $g(x) = x^2+x+1$

$f+g = 3x^2+4x+2$

$T(f+g) = 2x+3-2 = 2x+1$

$T(f)+T(g) = (1)x+2-2+(1)x+1-2 = 2x-1$

$\neq$

Ex 8: $T: M_{2,2} \longrightarrow \mathbb{R}$

$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 2a+2d$ Is T linear transf.

let $A_1 = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$, $A_2 = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$, $A_1+A_2 = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{bmatrix}$

$T(A_1)+T(A_2) = 2a_1+2d_1+2a_2+2d_2$ ✓ ①✓

$T(A_1+A_2) = 2(a_1+a_2)+2(d_1+d_2)$

$T(kA_1) = T\begin{bmatrix} ka_1 & kb_1 \\ kc_1 & kd_1 \end{bmatrix} = 2ka_1+2kd_1$ = ✓②✓

$kT(A) = k(2a_1+2d_1) = 2ka_1+2kd_1$

So T is linear transformation

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Ex: $T: R^2 \longrightarrow R^2$ is linear transformation

given $T(1,0) = (2,-2)$, $T(0,1) = (4,1)$

Find (a) $T(3,-2)$ (b) $T(a,b)$

(a) $T(3,-2) = T(3(1,0) - 2(0,1))$

$= 3T(1,0) - 2T(0,1)$

$= 3(2,-2) - 2(4,1)$

$= (-2, -8)$

(b) $T(a,b) = T(a(1,0) + b(0,1))$

$= aT(1,0) + bT(0,1)$

$= a(2,-2) + b(4,1)$

$= (2a+4b, -2a+b)$

Ex: $T: R^2 \longrightarrow R^3$ is linear transf.

Given $T(-2,1) = (-1,2,0)$, $T(1,3) = (0,-3,5)$

Find $T(x,y)$.

$T(-2,1) = -2T(1,0) + 1T(0,1) = (-1,2,0)$

$T(1,3) = 1.T(1,0) + 3T(0,1) = (0,-3,5)$

Let $T(1,0) = x$, $T(0,1) = y$ $\rightarrow$ $-2x + y = (-1,2,0)$

$x + 3y = (0,-3,5)$

Find $x, y \rightarrow x = T(1,0) = \left(\frac{-1}{7}, \frac{-4}{7}, \frac{10}{7}\right)$

$y = T(0,1) = \left(\frac{3}{7}, \frac{-9}{7}, \frac{5}{7}\right)$

$T(x,y) = xT(1,0) + yT(0,1) = x\left(\frac{-1}{7}, \frac{-4}{7}, \frac{10}{7}\right) + y\left(\frac{3}{7}, \frac{-9}{7}, \frac{5}{7}\right)$

$= \left(\frac{3x-y}{7}, \frac{-4x-9y}{7}, \frac{10x+5y}{7}\right)$.

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Ex: $T: P_1 \longrightarrow P_1$ is linear transf.

Given that $T(x-1) = x+2$, $T(x+2) = 2x+1$

find $T(3x+4)$

$3x+4 = 3(x) + 4(1)$

$T(3x+4) = 3T(x) + 4T(1)$

we need to find $T(x)$, $T(1)$

$T(x-1) = T(x) - T(1) = x+2$

$T(x+2) = T(x) + 2T(1) = 2x+1$

find $T(x)$, $T(1)$

$T(x) = \frac{4}{3}x + \frac{5}{3}$

$T(1) = \frac{1}{3}x - \frac{1}{3}$

$\rightarrow T(3x+4) = 3T(x) + 4T(1)$

$= 3\left(\frac{4}{3}x + \frac{5}{3}\right) + 4\left(\frac{1}{3}x - \frac{1}{3}\right)$

$= \frac{16}{3}x + \frac{11}{3}$

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The kernel and Range of linear transformation

Definition (kernel)

If $T: V \longrightarrow W$ is linear transformation

The kernel of T denoted by $\ker T$ is

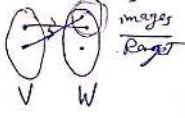
$$\ker T = \{ u \in V \ / \ T(u) = 0_W \}$$

Definition (Range)

If $T: V \longrightarrow W$ is linear transformation

, then the range of T denoted by $\text{Range}\, T$ is the set of all vectors in W that are images, i.e.

$$\text{Range}\, T = \{ w \in W \ / \text{ there is } u \in V \text{ such that } T(u) = w \}$$



Ex: $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^3$

$T(x,y) = (0,0,0)$ find $\ker T$ and $\text{Range}\, T$

$T(u) = 0_W$ for any $u \in V$

So $\ker T = \mathbb{R}^2$

and $\text{Range}\, T = (0,0,0)$.

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Ex: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ is linear transf.

$T(x,y,z) = (x,y)$

Find $\ker T$, Range T

$\ker T$ $u = (x,y,z)$ s.t. $T(u) = 0_W$

i.e. $T(x,y,z) = (0,0)$

$(x,y) = (0,0)$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$\Rightarrow x = y = 0$

$\Rightarrow u = (x,y,z) = (0,0,z)$

$\therefore \ker T = \{(0,0,z) \;/\; z \in \mathbb{R}\}$.

Range $T = \{(x,y) \;/\; x,y \in \mathbb{R}\} = \mathbb{R}^2$.

Ex: $T: \mathbb{R}^4 \longrightarrow \mathbb{R}^2$ Find $\ker T$

$T(x,y,z,w) = (x+y, z+w)$

Find $u = (x,y,z,w)$ s.t. $T(u) = 0_W$

$T(x,y,z,w) = (0,0)$

$(x+y, z+w) = (0,0) \longrightarrow \begin{matrix} x+y=0 \\ z+w=0 \end{matrix}$

$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ $(x,y,z,w) = (-t, t, -s, +s)$, $s,t \in \mathbb{R}$

$= \ker T$

$\ker T = \{(-t,t,-s,+s) \;/\; t,s \in \mathbb{R}\}$

$T(-t,t,-s,+s) = (-t+t, -s+s) = (0,0)$ ✓.

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Ex: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$T(x,y,z) = (x-y,\ x+2y,\ z)$

find $\ker T$

$\ker T = \{(x,y,z) \ / \ T(x,y,z) = 0_W\}$

$T(x,y,z) = (0,0,0)$

$(x-y,\ x+2y,\ z) = (0,0,0)$

$$\begin{aligned} x-y &= 0 \\ x+2y &= 0 \\ z &= 0 \end{aligned} \rightsquigarrow \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow[\text{row operations}]{\text{by elementary}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$\rightarrow \quad x = y = z = 0$

$\ker T = \{(0,0,0)\}$

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Theorem:

If $T: V \longrightarrow W$ is linear transformation, then

(a) $\ker T$ is subspace of V.

(b) $\text{Range } T$ is subspace of W.

Proof of (a)

(1) Let $u, v \in \ker T$ is $u+v \in \ker T$ (if $T(u+v) = 0_W$)

$$T(u+v) = T(u) + T(v) = 0_W + 0_W = 0_W$$

(T is linear transf. ↓; $u \in \ker T$, $v \in \ker T$ ↓)

So $u+v \in \ker T$

(2) Let $c \in \mathbb{R}$ $u \in \ker T$ is $cu \in \ker T$ (if $T(cu) = 0_W$)

$$T(cu) = cT(u) = c \cdot 0_W = 0_W$$

(T is linear transf ↓; $u \in \ker T$ ↓)

So $cu \in \ker T$

So $\ker T$ is subspace of V.

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* If $T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$

$T(u) = Au$ A is $m \times n$ matrix

Range $T \equiv$ column space of A.

basis for Range $T \equiv$ Basis for columnspace of A.

$\equiv$ " " row space of A^t.

ker $T \equiv$ Null space $\equiv$ solution space of A.

basis for ker $T \equiv$ basis for solution space $[A:0]$

dimension = number of vectors in any basis.

Theorem:

If $T: V \longrightarrow W$ is linear transformation

, then dim ker T + dim Range T = dim V

($\equiv$ nullity of A + rank A = no. of columns of A)

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Definition (one-one)

A linear transformation $T: V \to W$ is one-one if for all $u, v \in V$ and $T(u) = T(v)$, then $u = v$.

Theorem:

A linear transformation $T: V \to W$ is one-one if and only if $\ker T = \{0_V\}$

Definition (onto)

A linear transformation $T: V \to W$ is onto if for any $w \in W$ there is $u \in V$, $T(u) = w$ that is $\text{Range } T = W$.

* $\boxed{\dim \ker T + \dim \text{Range} T = \dim V}$
* If T is 1-1, then $\dim \ker T = 0$. and then $\dim \text{Range} T = \dim V$.
* If T is onto, then $\dim \text{Range} T = \dim W$ and then $\dim \ker T = \dim V - \dim W$.

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Ex: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$T(x,y,z) = (x+z,\ x+y+2z,\ 2x+y+3z)$

a) Is T onto, one-one?

b) Find basis for Range T and basis for $\ker T$.

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+z \\ x+y+2z \\ 2x+y+3z \end{bmatrix}_{3\times 1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}_{3\times 3} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3\times 1}$$

$(T(u) = Au)$, $m \times n$, $\dim W$, $\dim V$

a) Is T onto?

if Range $T = W$ $\Leftrightarrow$ ~~basis~~ $\dim$ for Range $T = \dim$ ~~basis~~ of $W = 3$

Find basis for Range T = basis of row space of A^t

$$A^t = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Basis for Range $T = \{(1,0,1), (0,1,1)\}$

$\dim$ Range $= 2 \neq \dim W = 3$

So T is not onto.

is T one-one (if $\dim \ker T = 0$, then T is 1-1)

$\dim \ker T + \dim$ Range $T = \dim$ W

$\dim \ker T + 2 = 3$

$\rightarrow \dim \ker T = 1$

$\rightarrow T$ is not one-one.

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(b) basis for Range T

$= \{(1,0,1), (0,1,1)\}$

basis for ker T

= basis for solution space of A.

$$[A:0] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\ker T = \left\{ \begin{bmatrix} -t \\ -t \\ t \end{bmatrix} / t \in \mathbb{R} \right\} = \text{solution space}$$

$$\text{basis} = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Ex: $T: \mathbb{R}^4 \longrightarrow \mathbb{R}^3$

$T(a_1, a_2, a_3, a_4) = (a_1 + a_2, a_3 + a_4, a_1 + a_3)$

a) Find basis for Range T and ker T

b) Is T one-one ? onto ?

$$T\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \underset{3\times 1}{\begin{bmatrix} a_1+a_2 \\ a_3+a_4 \\ a_1+a_3 \end{bmatrix}} = \underset{3\times 4}{\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}} \underset{4\times 1}{\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}}$$

A is 3×4

↓ dim W ↘ dim V

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basis for range T

$$A^t = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Basis for Range $T = \{(1,0,0), (0,1,0), (0,0,1)\}$

dim Range $T = 3$

basis for ker T

$$[A:0] = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\ker T = \left\{ \begin{bmatrix} t \\ -t \\ -t \\ t \end{bmatrix} / t \in \mathbb{R} \right\}$$

basis for $\ker T = \left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\}$

dim $\ker T = 1$

b) dim $\ker T = 1 \neq 0 \rightarrow T$ is not 1-1

dim Range $T = 3 = \dim W$

$\rightarrow T$ is onto.

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Ex: $T: \mathbb{R}^4 \longrightarrow \mathbb{R}^3$

$T(x,y,z,w) = (x+y,\ y-z,\ z-w)$

a) Is $(1,-1,1,-1) \in \ker T$

b) Is $(2,-1,3) \in \text{Range } T$

a) $u \in \ker T$ if $T(u) = 0_W$.

So $T(1,-1,1,-1) = (0,-2,2) \neq (0,0,0)$

So $(1,-1,1,-1) \notin \ker T$

b) $w \in \text{Range } T$ if there is $u \in V$ $T(u) = w$

is there (x,y,z,w) $T(x,y,z,w) = (2,-1,3)$

find x,y,z,w $T(x,y,z,w) = (2,-1,3)$

$(x+y,\ y-z,\ z-w) = (2,-1,3)$

$x+y=2$
$y-z=-1$
$z-w=3$

Solve if there is solution unique or infinite then $w \in \text{Range } T$

If No sol^n $w \notin \text{Range } T$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & -1 & 3 \end{bmatrix}$$

infinite no. of solutions

So $(2,-1,3) \in \text{Range } T$.

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Ex: $T: P_3 \to W$ is linear transf.

T is one-one and onto

find $\dim W$

$\dim \ker T + \dim \text{Range } T = \dim V$

T is onto so $\dim \text{Range } T = \dim W$

T is one-one $\to$ $\dim \ker T = 0$

$\dim W = \dim V = 4$.

Ex: $T: P_n \to \mathbb{R}^3$ is linear transf.

T is onto $\dim \ker T = 2$ find n.

$\dim \ker T + \dim \text{Range } T = \dim V$

$2 + \dim W = n+1$

$2 + 3 = n+1$

$n = 4$

$V = P_4$.

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(8.3) Compositions and inverse transformations

Def: If $T_1: U \to V$, $T_2: V \to W$ are linear transf. then composition of T_2 with T_1 denoted by $T_2 \circ T_1$ is

$$T_2 \circ T_1(u) = T_2(T_1(u)).$$

Theorem:
If $T_1: U \to V$ and $T_2: V \to W$ are linear transf., then $T_2 \circ T_1: U \to W$ is linear transf.

Proof:

1) $T_2 \circ T_1(u_1+u_2) = T_2(T_1(u_1+u_2)) \underset{T_1 \text{ linear}}{=} T_2(T_1(u_1)+T_1(u_2))$

$\underset{T_2 \text{ linear}}{=} T_2(T_1(u_1)) + T_2(T_1(u_2)) = T_2 \circ T_1(u_1) + T_2 \circ T_1(u_2)$

2) $T_2 \circ T_1(ku) = T_2(T_1(ku)) \underset{T_1 \text{ linear}}{=} T_2(kT_1(u)) \underset{T_2 \text{ linear}}{=} kT_2(T_1(u))$

$= k\, T_2 \circ T_1(u)$

$\Rightarrow T_2 \circ T_1$ is linear transf.

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Ex 1: Let $T_1 : P_1 \to P_2$, $T_2 : P_2 \to P_2$ linear trasf.

$T_1(p(x)) = x p(x)$ and $T_2(p(x)) = p(2x+4)$

find (a) $T_2 \circ T_1 (p(x))$ (b) $T_2 \circ T_1 (3x-4)$

(a) $T_2 \circ T_1 (p(x)) = T_2(T_1(p(x))) = T_2(x\,p(x))$

$= (2x+4)\,p(2x+4)$

(b) $T_2 \circ T_1 (3x-4) = (2x+4)(3x-4)$

$= (2x+4)(3(2x+4)-4)$

$= (2x+4)(6x+12-4) = (2x+4)(6x+8)$

$= 12x^2 + 40x + 32$

Ex 2: $T_1 : R^2 \to R^2$ $\quad T_1(x,y) = (2x, 3y)$

$T_2 : R^2 \to R^2$ $\quad T_2(x,y) = (x-y, x+y)$

find (a) $T_2 \circ T_1 (x,y)$ (b) $T_2 \circ T_1 (3,2)$

(a) $T_2 \circ T_1 (x,y) = T_2(T_1(x,y)) = T_2(2x, 3y)$

$= (2x-3y, 2x+3y)$

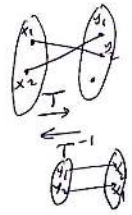
(b) $T_2 \circ T_1 (3,2) = (0, 12)$

or $T_2(T_1(3,2)) = T_2(6,6) = (0, 12)$

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Inverse linear transformation

If $T: V \longrightarrow W$ is one-one linear transf., then T^{-1} exist $T^{-1}: R(T) \xrightarrow{\subseteq} V$



Ex: $T: P_1 \longrightarrow P_2$

$T(p(x)) = x\,p(x)$ find T^{-1}

$T(a+bx) = ax + bx^2$

$T^{-1}: R(T) \xrightarrow{\subseteq} P_1$

$T^{-1}(ax+bx^2) = a+bx$

$T^{-1}(2x-3x^2) = 2-3x$

Ex: $T: R^3 \longrightarrow R^3$ $T(x_1,x_2,x_3) = (3x_1+x_2, -2x_1-4x_2+3x_3, 5x_1+4x_2-2x_3)$

a) Determine whether T is one-one

b) Find $T^{-1}(x_1,x_2,x_3)$

$$T\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$T(X) = AX$

$\det A = 3(8-12) - 1(4-15) = -12+11 = -1 \neq 0$

$\det A \neq 0 \longrightarrow A$ invert. $\longrightarrow$ sol: $AX=0$ is only trivial

$\longrightarrow \ker T = \{(0,0,0)\} \longrightarrow T$ is 1-1

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$\Rightarrow T^{-1}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = A^{-1}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -12 & 7 & 10 \end{bmatrix}$$

$$\Rightarrow T^{-1}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -12 & 7 & 10 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T^{-1}(x_1, x_2, x_3) = (4x_1 - 2x_2 - 3x_3,\ -11x_1 + 6x_2 + 9x_3,\ -12x_1 + 7x_2 + 10x_3)$$

Ex: $T: R^3 \longrightarrow R^3$

$T(x_1, x_2, x_3) = (x_1 + 5x_2 + 2x_3,\ x_1 + 2x_2 + x_3,\ -x_1 + x_2)$

Determine whether T has inverse or not

$$T\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 2 \\ 1 & 2 & 1 \\ -1 & 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\det A = 0 \rightarrow A$ not invertible

$\rightarrow$ Sol^n of $AX = 0$ non trivial

$\rightarrow \ker T \neq \{0_V\}$

$\rightarrow T$ is not 1-1

$\rightarrow T$ has No inverse

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